

Scoring Contingent Auctions for Procurement: Theory and Experimental Evidence*

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Abstract

We theoretically and experimentally study procurement auctions for infrastructure projects with post-auction risk. We compare fixed-payment (cash) auctions, scaling (unit-price) auctions, and a novel cost-sharing auction in which bidders compete by requesting a fraction of variable costs to be covered. Our experiments vary how bidder heterogeneity enters the environment: through input-quantity risk or through marginal costs. We find that scaling auctions can outperform fixed-payment auctions when heterogeneity operates through marginal costs but perform poorly when heterogeneity operates through quantity risk. Our theoretical analysis of cost-sharing auctions demonstrates how they can combine efficient allocation with risk sharing. Our results provide experimental evidence on the trade-off between risk sharing and adverse selection in procurement design.

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1 Introduction

Public infrastructure projects such as highways, bridges, and rail systems are commonly allocated through procurement auctions. Yet substantial uncertainty often remains when contracts are awarded, so procurement design must determine not only which firm performs the project, but also how project risk is shared between the buyer and the contractor. Given the scale of public procurement, this risk-allocation problem has important fiscal consequences.

In practice, procurement agencies address this problem through different combinations of fixed and contingent payments. Fixed-payment (cash) auctions leave the contractor bearing the full project risk, whereas scaling (unit-price) auctions condition payments on realized input quantities, thereby shifting part of this risk to the buyer.¹ We also consider cost-sharing contracts, which reimburse a fraction of realized project costs and thus provide an alternative form of contingent payment, albeit one that requires the buyer to observe realized costs.

The relative performance of these mechanisms reflects three forces. First, the payment rule affects which firms bid most aggressively and hence the allocation of the project. Second, contingent payments provide insurance by shifting project risk to the buyer. Third, cost sharing creates a linkage between payments and realized costs that can reduce informational rents. Our analysis therefore identifies when the commonly used fixed-payment and scaling formats are preferable and evaluates the potential benefits—and informational requirements—of cost sharing as an alternative.

To separate these forces empirically, we experimentally compare the three mechanisms

¹In a scaling auction, the project is partitioned into tasks, each requiring a potentially uncertain quantity of inputs. Firms bid a unit price for each task, and their score is a weighted average of these prices. The contract is awarded to the lowest-scoring bidder, who is paid its bid prices for the quantities actually realized.

across environments that vary the source of bidder heterogeneity and the structure of quantity risk. In one environment, firms differ in marginal costs while facing common quantity uncertainty; in the other, they differ in input requirements while sharing common marginal costs. This variation allows us to disentangle how the mechanism affects allocation, reallocates project risk, and links payments to realized costs. Such comparisons are difficult in observational data: procurement agencies rarely employ multiple mechanisms in otherwise comparable settings, projects differ substantially in scale and complexity, and the underlying source of firm heterogeneity—whether in costs, input requirements, or exposure to risk—is typically difficult to identify separately.

Our experiment yields three main findings. First, the performance of scaling auctions depends critically on the source of bidder heterogeneity. When firms differ in input quantities, scaling auctions can induce perverse selection: less efficient firms bid more aggressively and are more likely to win. Cost-sharing auctions restore the efficient ordering of bids, substantially improving allocative efficiency and reducing procurement costs.

Second, when firms instead differ in marginal costs, the performance of scaling auctions depends on how quantity shocks are correlated across tasks. Scaling auctions increase procurement costs relative to fixed-payment auctions when shocks are positively correlated, but reduce them when shocks are negatively correlated. This difference is consistent with the risk-allocation properties of scaling: negatively correlated quantities allow bidders to hedge cost risk through their unit-price bids, whereas positively correlated quantities leave them substantially exposed to risk.

Third, behavioral departures from risk-neutral bidding vary systematically across mechanisms. Bidding in fixed-payment and cost-sharing auctions is consistent with relatively modest loss aversion, whereas scaling under positively correlated quantity shocks is consis-

tent with substantially stronger loss aversion. Under negative correlation, bidders instead use their unit-price bids to partially hedge project risk. Thus, the behavioral response to contingent payments depends critically on the risk they leave bidders bearing.

Our results collectively show that the performance of contingent procurement mechanisms cannot be assessed from their insurance properties alone. Their effects depend jointly on which firms they induce to win, how they allocate project risk, and how strongly payments are linked to realized costs. This helps explain why the same contingent-payment mechanism can perform well in one procurement environment and poorly in another.

Related literature Our paper contributes to three related literatures: scoring auctions, contingent-payment procurement, and procurement experiments. Together, these literatures motivate our analysis of how alternative contingent-payment mechanisms perform under different sources of seller heterogeneity.

Seminal contributions by [Che \(1993\)](#) and [Asker and Cantillon \(2008, 2010\)](#) characterize equilibrium behavior and optimal scoring rules in deterministic scoring auctions under asymmetric information. More recent work extends these results to richer procurement environments and nonlinear scoring rules ([Aways et al., 2025](#); [Hanazono et al., 2024](#)).

While deterministic scoring auctions provide a useful benchmark, many procurement settings involve uncertainty over project outcomes, giving rise to contingent scoring auctions. [Ewerhart and Fieseler \(2003\)](#) provide a foundational analysis of such mechanisms in a procurement problem with a fixed and a contingent task. In their framework, heterogeneity operates through the realized labor requirement of the contingent task rather than through private cost parameters. They show that equilibrium bidding may be nonmonotone and allocations inefficient, while contingent payments can intensify competition and reduce

informational rents.

More recent theoretical work studies multidimensional procurement with stochastic input quantities and risk-averse sellers. In contrast to [Ewerhart and Fieseler \(2003\)](#), where heterogeneity arises through the realized labor requirement of a single contingent task, [Hernández-Chanto and Rivera Mora \(2025\)](#) characterize optimal scaling auctions in environments with multiple contingent tasks and seller heterogeneity in fixed costs. They show that weights proportional to expected quantities are optimal, provide a simple rule for when scaling auctions outperform conventional cash auctions based on the relative importance of marginal and fixed costs, and establish that hybrid auctions combining contingent and lump-sum payments outperform both scaling and cash auctions.

Beyond fixed-payment and scaling auctions, recent work studies procurement mechanisms that condition payments on realized costs. [Bergemann et al. \(2025\)](#) study robust procurement without priors and show that simple contingent-payment mechanisms sharing a fixed fraction of buyer utility with the seller can perform optimally in a worst-case sense.

The experimental literature has likewise examined a broad range of procurement mechanisms, from classical deterministic scoring auctions to more recent environments with contingent payments. A recurring finding is that bidder behavior often departs from equilibrium predictions, affecting the relative performance of alternative procurement designs.

Several experimental studies have examined deterministic scoring auctions and other multiattribute procurement mechanisms; see, among others, [Bichler \(2000\)](#), [Chen-Ritzo et al. \(2005\)](#), [Strecker \(2010\)](#), and [Albano et al. \(2024\)](#). Building on this literature, [Camboni et al. \(2025\)](#) compare one-dimensional procurement auctions with scoring auctions that allow firms to compete simultaneously over price and quality. While theory predicts that the richer bidding space of scoring auctions should improve buyer welfare, the experimental evidence is

more nuanced. Bidding becomes substantially noisier as the strategy space expands, largely eliminating the theoretical advantage of scoring auctions. The authors attribute this finding to the cognitive complexity of multidimensional bidding.

Recent research also examines procurement mechanisms with contingent payments. [Shachat et al. \(2024\)](#) show experimentally that moderate reliance on contingent payments improves procurement outcomes, whereas excessively weak or strong incentives reduce performance. Their setting focuses on quality provision and moral hazard, whereas we abstract from moral hazard to study the risk-allocation properties of alternative contingent-payment schemes.

Experimental evidence on scaling auctions remains largely absent. Existing evidence comes primarily from structural and observational studies of individual procurement markets. [Bolotnyy and Vasserman \(2023\)](#) show that scaling contracts provide partial insurance against quantity risk but leave sellers exposed to residual uncertainty, generating risk premia and bid skewing. [Luo and Takahashi \(2025\)](#) likewise show that contractual arrangements affect participation, bidding, and risk allocation. In turn, [Fok \(2023\)](#) emphasizes the value of information acquisition under project uncertainty. While these studies provide rich evidence from particular procurement markets, the source of bidder heterogeneity and the contractual environment are difficult to identify separately. Our experiment complements this literature by varying these dimensions directly and isolating the allocation, insurance, and linkage forces that shape the performance of contingent procurement mechanisms.

Finally, our paper contributes to the growing experimental literature on auctions with contingent claims. In procurement, [Bos et al. \(2025\)](#) study profit-sharing auctions in which suppliers bid the share of project profits they require rather than a fixed payment. They find that profit sharing lowers buyer expenditures relative to cash auctions, although the gains

are smaller than predicted because of noisy bidding and residual inefficiencies. More broadly, our work relates to the literature on experimental security-bid auctions. Breig et al. (2026), Breig et al. (2025), and Bajoori et al. (2024) study auctions in which bidders compete using securities such as debt and equity, showing how the shape of the contingent claim affects bidding incentives and auction performance.

Organization of the paper The remainder of the paper is organized as follows. Section 2 introduces the model and the three procurement auction formats. Section 3 characterizes equilibrium behavior and compares allocative efficiency and procurement costs across mechanisms. Section 4 describes the experimental design, and Section 5 presents the experimental results. Section 6 concludes. All auxiliary theoretical results and proofs are collected in the Appendix.

2 Model

Consider a buyer who wishes to procure a project. The project consists of a set of tasks $\mathcal{T} = \{1, \dots, T\}$, all of which must be completed for its successful execution. There is a set $\mathcal{I} = \{1, \dots, I\}$ of potential sellers, with $I \geq 2$.

Prior to the auction, each seller i privately learns a type $\theta_i \in \Theta := [\underline{\theta}, \bar{\theta}]$, drawn independently from a commonly known, absolutely continuous distribution F with a continuous density f that is strictly positive on Θ . A seller's type determines three primitives: a base fixed cost $k(\theta)$, a deterministic base input requirement $\mathbf{q}(\theta)$, and a base marginal cost vector $\mathbf{c}(\theta) = (c_t(\theta) : t \in \mathcal{T})$.

Actual input requirements and production costs are uncertain. Specifically, the total quantity of inputs required from a seller of type θ is $\mathbf{q}(\theta) + \boldsymbol{\xi}^q$, where $\boldsymbol{\xi}^q = (\xi_t^q : t \in \mathcal{T})$;

realized marginal costs are $\mathbf{c}(\theta) + \boldsymbol{\xi}^c$, where $\boldsymbol{\xi}^c = (\xi_t^c : t \in \mathcal{T})$; and realized fixed costs are $k(\theta) + \xi^k$. The vector of random shocks $\boldsymbol{\xi} = (\boldsymbol{\xi}^q, \boldsymbol{\xi}^c, \xi^k)$ is common to all sellers, independent of their types, and drawn from a commonly known distribution $\nu \in \Delta(\mathbb{R}_+^T \times \mathbb{R}_+^T \times \mathbb{R}_+)$ with bounded support.² The shocks are realized after the auction selects the winning seller.

The mappings $k(\cdot)$, $\mathbf{q}(\cdot)$, and $\mathbf{c}(\cdot)$ are differentiable. Fixed costs $k(\theta) + \xi^k$ and variable costs $(\mathbf{q}(\theta) + \boldsymbol{\xi}^q) \cdot (\mathbf{c}(\theta) + \boldsymbol{\xi}^c)$ are assumed to be weakly increasing in θ for each realization $\boldsymbol{\xi}$. Total cost, $k(\theta) + \xi^k + (\mathbf{q}(\theta) + \boldsymbol{\xi}^q) \cdot (\mathbf{c}(\theta) + \boldsymbol{\xi}^c)$, is assumed to be strictly increasing in θ for each realization $\boldsymbol{\xi}$. Thus, in this sense, the type θ is an index of (in)efficiency.

We say that heterogeneity operates exclusively through fixed costs if $\mathbf{q}(\theta) \equiv \mathbf{q}$ and $\mathbf{c}(\theta) \equiv \mathbf{c}$; exclusively through marginal costs if $k(\theta) \equiv k$ and $\mathbf{q}(\theta) \equiv \mathbf{q}$; and exclusively through input quantities if $k(\theta) \equiv k$ and $\mathbf{c}(\theta) \equiv \mathbf{c}$, where k , $\mathbf{q}$, and $\mathbf{c}$ denote the corresponding deterministic baselines.

Sellers value their final monetary outcome using a Bernoulli utility function $u : \mathbb{R} \rightarrow \mathbb{R}$ evaluated at the auction's revenue minus the costs of delivering the project. The utility u is strictly increasing, twice differentiable, weakly concave, and satisfies $u(0) = 0$. The buyer, on the other hand, is risk neutral and aims to minimize expected payment.

An *environment* is a tuple $\mathcal{E} = (\mathcal{I}, [\underline{\theta}, \bar{\theta}], F, k(\cdot), \mathbf{q}(\cdot), \mathbf{c}(\cdot), u(\cdot), \nu)$ that fully describes the number of sellers, their cost structure, their risk preferences, and the common stochastic components.

2.1 Scoring contingent auctions

We study a broad class of scoring auctions for allocating the project. Sellers submit bids summarized by a score s , and the project is awarded to the seller submitting the lowest

²For notational simplicity, random variables and vectors and their realizations are denoted by the same symbols.

score. The winning seller's payment depends on the score and on a vector of contractable observables $\mathbf{O}(\theta, \boldsymbol{\xi})$.³

A scoring auction, denoted by $\mathcal{A}$, consists of three components: a nonempty compact and convex set of permissible scores $\mathcal{S} \subseteq \mathbb{R}$; a compact-valued correspondence of alternatives $\{\mathcal{B}(s)\}_{s \in \mathcal{S}}$; and a continuous payment rule $\Pi(\cdot)$ that specifies the buyer's contingent payment to the winning seller as a function of the contractable observables $\mathbf{O}(\theta, \boldsymbol{\xi})$, the score s , and the selected alternative $b \in \mathcal{B}(s)$.

The procurement process unfolds as follows. Sellers privately observe their types and simultaneously submit bids. Seller i 's bid consists of a score $s_i \in \mathcal{S}$ and an associated alternative $b_i \in \mathcal{B}(s_i)$. The seller submitting the lowest score is awarded the project.⁴ If seller i wins, it undertakes the project, incurring fixed cost $k(\theta_i) + \xi^k$ and variable cost $(\mathbf{q}(\theta_i) + \boldsymbol{\xi}^q) \cdot (\mathbf{c}(\theta_i) + \boldsymbol{\xi}^c)$. The buyer compensates seller i according to the payment rule $\Pi(\mathbf{O}(\theta_i, \boldsymbol{\xi}), s_i, b_i)$. Thus, seller i 's ex-post monetary payoff conditional on winning is

$$\pi_i(\theta_i, s_i, b_i, \boldsymbol{\xi}) := \Pi(\mathbf{O}(\theta_i, \boldsymbol{\xi}), s_i, b_i) - (\mathbf{q}(\theta_i) + \boldsymbol{\xi}^q) \cdot (\mathbf{c}(\theta_i) + \boldsymbol{\xi}^c) - k(\theta_i) - \xi^k.$$

The interim expected utility of seller i with type θ_i , conditional on winning with score s_i in auction $\mathcal{A}$, is

$$U(s_i, \theta_i \mid \mathcal{A}) := \max_{b_i \in \mathcal{B}(s_i)} \mathbb{E}[u(\pi_i(\theta_i, s_i, b_i, \boldsymbol{\xi}))],$$

where the expectation is taken with respect to the vector of shocks $\boldsymbol{\xi}$. We assume that $U(s, \theta \mid \mathcal{A})$ is strictly increasing in s , so that, conditional on winning, every seller type strictly prefers a higher score.

³The buyer observes $\mathbf{O}(\theta, \boldsymbol{\xi})$ without observing θ or $\boldsymbol{\xi}$. For instance, Departments of Transportation typically employ independent engineers or inspectors to verify the quantities of materials used. Some contracts additionally verify realized costs through on-site inspections, material testing, and detailed record-keeping.

⁴Ties are broken uniformly at random.

Timing. The environment unfolds in four stages: $t = 0$ (ex ante), sellers have not yet observed their types; $t = 1$ (interim), sellers privately observe their types and simultaneously submit bids; $t = 2$ (post award), the project is awarded to the seller submitting the lowest score and the winner learns it has obtained the contract, but quantity uncertainty has not yet been realized; and $t = 3$ (ex post), the vector of shocks $\boldsymbol{\xi}$ is realized, implementation takes place, payments are determined, and payoffs are received. Whenever convenient, we refer to information sets by their corresponding time index.

2.2 Auction formats

Throughout the paper, we focus on the three auction formats described below.

Fixed-payment auctions In a *fixed-payment auction* ($\mathcal{A}_{FP}$), each bidder i competes by submitting a fixed payment s_i that does not depend on observables. If bidder i wins, the payment rule is simply $\Pi(\mathbf{O}(\theta, \boldsymbol{\xi}), s_i, b_i) = s_i$.⁵

Scaling auctions A *scaling auction* ($\mathcal{A}_S$) is a specific type of scoring auction in which the contractable observables are the realized input quantities, $\mathbf{O}(\theta, \boldsymbol{\xi}) = \mathbf{q}(\theta) + \boldsymbol{\xi}^q$, and sellers bid a vector of per-unit prices $\mathbf{b}$, leading to payment $\mathbf{O}(\theta, \boldsymbol{\xi}) \cdot \mathbf{b}$. A scoring rule is defined by a vector $\mathbf{w} = (w_t : t \in \mathcal{T}) \in \mathbb{R}_{++}^T$ of task-specific weights, which the buyer determines and announces at the outset of the auction. Each seller i observes the scoring rule and submits a non-negative per-unit bid $b_{i,t} \geq 0$ for every task t . All bid vectors $\mathbf{b}_i \in \mathbb{R}_+^T$ are submitted simultaneously. Each vector $\mathbf{b}_i$ induces a score $s_i = \mathbf{b}_i \cdot \mathbf{w}$, with permissible scores lying in $\mathcal{S}$. The contract is awarded to the seller submitting the *lowest score*, who then executes the

⁵For each s_i , the set of alternatives $\mathcal{B}(s_i)$ contains only s_i .

project.⁶

Cost-sharing auctions The class of scoring auctions is broad enough to encompass certain security-bid auctions as special cases. In particular, one can construct auctions in which bids specify a security written on realized variable cost, $\mathbf{O}(\theta, \boldsymbol{\xi}) = (\mathbf{q}(\theta_i) + \boldsymbol{\xi}^q) \cdot (\mathbf{c}(\theta_i) + \boldsymbol{\xi}^c)$. A particularly simple example is a *cost-sharing auction* ($\mathcal{A}_{CS}$), in which the score space is $\mathcal{S} = [0, 1]$.⁷ Each seller i submits a bid $s_i \in [0, 1]$, interpreted as the fraction of realized variable cost that the seller requests to be reimbursed. The winning seller receives a fixed lump-sum payment X plus reimbursement of the fraction s_i of realized variable cost, where $\mathbb{E}[u(X - k(\bar{\theta}) - \xi^k)] > 0$. Thus, the buyer's payment to the winning seller is $\Pi(\mathbf{O}(\theta, \boldsymbol{\xi}), s) = X + s_i \mathbf{O}(\theta_i, \boldsymbol{\xi})$.

3 Equilibrium Analysis

Fix an environment $\mathcal{E}$ and a scoring auction $\mathcal{A}$. Together, the pair $(\mathcal{E}, \mathcal{A})$ induces a Bayesian game in which each firm privately observes its own type and chooses a score $s \in \mathcal{S}$ together with an associated alternative $b \in \mathcal{B}(s)$. We are interested in characterizing equilibrium behavior in the Bayesian games induced by different scoring auctions.

Definition 1. *An auction $\mathcal{A}$ is rich for environment $\mathcal{E}$ if, for each type θ , there exists a score $s \in \mathcal{S}$ such that $U(s, \theta \mid \mathcal{A}) = 0$.*

⁶To see that this auction fits within the scoring-auction framework, note that each seller i submits a bid vector $\mathbf{b}_i \in \mathbb{R}_+^T$, which serves as the chosen “alternative.” The score $s_i = \mathbf{b}_i \cdot \mathbf{w}$ is derived from this bid. For a given score s , the available alternatives are $\mathcal{B}(s) = \{\mathbf{b} \in \mathbb{R}_+^T \mid \mathbf{b} \cdot \mathbf{w} = s\}$. The winning seller i receives total payment $\Pi(\mathbf{O}, s_i, \mathbf{b}_i) = \mathbf{O} \cdot \mathbf{b}_i$.

⁷More generally, the buyer could use other standard securities written on realized variable cost. For example, a call option on $\mathbf{O}$ is equivalent to an insurance contract with a deductible, while a debt-like security corresponds to reimbursement capped at a fixed amount.

This definition implies that the available set of scores $\mathcal{S}$ covers a range broad enough to satisfy two conditions for every type θ : (i) participation is individually rational; and (ii) the buyer can extract all surplus under sufficient competition.⁸

Let $\varphi_{\mathcal{A}} : [\underline{\theta}, \bar{\theta}] \rightarrow \mathbb{R}$ denote a bidding strategy in auction $\mathcal{A}$, and suppose that all sellers other than i bid according to $\varphi_{\mathcal{A}}$. Given a seller of type θ_i , the expected utility from submitting score s_i is

$$\text{EU}(s_i \mid \theta_i, \mathcal{A}, \varphi_{\mathcal{A}}) := U(s_i, \theta_i \mid \mathcal{A}) \cdot \Pr[s_i < \varphi_{\mathcal{A}}(\theta_j) \text{ for every } j \neq i].$$

Definition 2. Fix an auction mechanism $\mathcal{A}$. A score-bidding function $\varphi_{\mathcal{A}}$ is a symmetric Bayesian equilibrium of $\mathcal{A}$ if, for each bidder $i \in \mathcal{I}$, each type $\theta_i \in [\underline{\theta}, \bar{\theta}]$, and each feasible score $s_i \in \mathcal{S}$,

$$\text{EU}(\varphi_{\mathcal{A}}(\theta_i) \mid \theta_i, \mathcal{A}, \varphi_{\mathcal{A}}) \geq \text{EU}(s_i \mid \theta_i, \mathcal{A}, \varphi_{\mathcal{A}}).$$

3.1 Efficiency

This section relates project allocation to the source of firms' heterogeneity. Any type that obtains positive equilibrium utility must choose a score in $\Pi_+^{\mathcal{A}} := \{(\theta, s) : U(s, \theta \mid \mathcal{A}) > 0\}$, the set of type-score pairs that yield positive utility.

Definition 3. Fix an environment $\mathcal{E}$ and an auction $\mathcal{A}$ that is rich for $\mathcal{E}$. Suppose $U(s, \theta \mid \mathcal{A})$ is jointly continuous on $S \times \Theta$ and differentiable in s , with a jointly continuous, strictly

⁸While mechanisms like reserve prices can reduce the government's expected payment, they introduce a positive probability that no firm procures the project. Condition (i) ensures the project is always allocated, a requirement often necessary for the provision of public projects. Moreover, the existence of a score s such that $U(s, \theta \mid \mathcal{A}) = 0$ also implies condition (ii): if the number of competitor firms is large enough, competition is able to push firms to bid the lowest score they can profitably offer, namely a score that provides zero utility. In this sense, rich auctions are able to extract all surplus, provided that competition is sufficiently large.

positive score derivative $U_1(s, \theta \mid \mathcal{A})$.

1. $\mathcal{A}$ is **well behaved** if U is strictly log-supermodular on Π_+^A and, for every θ' and $s \in S$ with $U(s, \theta' \mid \mathcal{A}) = 0$, $U(s, \theta \mid \mathcal{A}) > 0$ for all $\theta < \theta'$.
2. $\mathcal{A}$ **induces perverse incentives** if U is strictly log-submodular on Π_+^A and, for every θ' and $s \in S$ with $U(s, \theta' \mid \mathcal{A}) = 0$, $U(s, \theta \mid \mathcal{A}) > 0$ for all $\theta > \theta'$.

Definition 3 classifies auction formats according to how type affects a seller's incentive to bid aggressively. In a well-behaved auction, more efficient types have stronger incentives to submit lower scores, leading to an equilibrium bidding strategy that is increasing in type. Under perverse incentives, the opposite holds: less efficient types have stronger incentives to submit lower scores, leading to a decreasing equilibrium bidding strategy.

Proposition 1 in the Appendix formalizes these implications. If $\mathcal{A}$ is well behaved for $\mathcal{E}$, the symmetric equilibrium is increasing in θ , and the most efficient firm wins. If instead $\mathcal{A}$ induces perverse incentives, the symmetric equilibrium is decreasing in θ , and the most inefficient firm wins.

Task elasticity We now proceed to classify which formats are well behaved. With this in mind, write $EC(\theta) := \mathbb{E}[(\mathbf{q}(\theta) + \boldsymbol{\xi}^q) \cdot (\mathbf{c}(\theta) + \boldsymbol{\xi}^c) + k(\theta) + \xi^k]$ for the expected total cost of type θ . A task t is *elastic* if

$$\frac{q'_t(\theta)}{q_t(\theta) + \mathbb{E}[\xi_t^q]} > \frac{EC'(\theta)}{EC(\theta)} \quad (1)$$

for every θ , and *inelastic* if the inequality is reversed. Thus, elasticity means that expected task- t inputs vary proportionally more than expected total costs. An environment is *elastic* (*inelastic*) if all tasks are elastic (inelastic).

Theorem 1. *Consider a rich auction environment.*

1. *The fixed payment auction and each cost-sharing auction are well-behaved.*
2. *Each scaling auction induces perverse incentives in each elastic risk-neutral environment and is well behaved in each inelastic risk-neutral environment.*

Remark 1. *Environments in which seller heterogeneity enters only through fixed or marginal costs are inelastic by definition. Hence, scaling auctions are well behaved in risk-neutral environments with these types of heterogeneity.*

In fixed-payment and cost-sharing auctions, the induced equilibrium is well behaved and therefore allocatively efficient across all environments.⁹ For scaling auctions, by contrast, efficiency depends on sellers' risk preferences and the elasticity of the tasks.¹⁰ In a risk-neutral elastic environment, a “markup-scaling” effect favors sellers with larger input requirements: because their quantities increase proportionally faster than their expected total costs, a given profitable score generates greater utility for higher types. Less efficient sellers consequently have stronger incentives to submit lower scores, reversing the equilibrium ordering and inducing a perverse equilibrium in which the least efficient seller is awarded the project. In a risk-neutral inelastic environment, by contrast, this effect is not strong enough to reverse the ordering of types; the equilibrium remains well behaved and scaling is allocatively efficient.

Remark 2. *Ewerhart and Fieseler (2003) provide an example with homogeneous costs and $T = 2$ tasks: an elastic task (labor) and an inelastic task (input material). While a scaling auction with low weights on the elastic task induces inefficient equilibria, a low weight on the*

⁹Cost-sharing auctions are rich only if $X \geq k(\bar{\theta})$. The payment X could be lower as long as bids above $s \geq 1$ are allowed. Nevertheless, this would lead the cost-sharing auction to induce perverse incentives.

¹⁰Theorem 1 can be generalized to settings in which not all tasks are elastic or inelastic. In particular, Part 2 continues to hold when the scoring rule assigns sufficiently low weights to inelastic (resp. elastic) tasks.

inelastic task induces efficient equilibria. For intermediate weights the equilibrium bidding strategy has an inverted U shape.

3.2 Buyer Expected Costs

Let $\mathbb{E}[C^{\mathcal{A}}]$ denote the buyer's ex-ante expected procurement cost under mechanism $\mathcal{A}$, evaluated at the symmetric equilibrium. We show that there are two forces that determine the cost ranking across auction formats. The first is allocative efficiency: when equilibrium scores are increasing in type, the project is assigned to the lowest-cost seller, while a scaling auction with perverse incentives assigns the project to the least efficient seller. (Lemma 4 identifies the resulting extra cost on the buyer implied by the allocation effect.) The second is a linkage effect. Holding the allocation fixed, mechanisms that make the winning seller's utility less sensitive to type reduce informational rents and lower expected procurement costs. (Lemma 5 formalizes this comparison by ranking well-behaved mechanisms in terms of their marginal effect of type on winning utility, $U_{\theta}(\varphi_{\mathcal{A}}(\theta), \theta \mid \mathcal{A})$.)

The next theorem summarizes how these two forces determine the ranking of expected procurement costs in the risk-neutral environments.

Theorem 2. *Consider a rich auction environment and suppose sellers are risk neutral.*

1. *If heterogeneity operates exclusively through fixed costs, then expected procurement costs satisfy: $\mathbb{E}[C^{\mathcal{A}_S}] = \mathbb{E}[C^{\mathcal{A}_{FP}}] = \mathbb{E}[C^{\mathcal{A}_{CS}}]$.*
2. *If heterogeneity operates exclusively through marginal costs, then expected procurement costs satisfy: $\mathbb{E}[C^{\mathcal{A}_S}] = \mathbb{E}[C^{\mathcal{A}_{FP}}] > \mathbb{E}[C^{\mathcal{A}_{CS}}]$.*
3. *If heterogeneity operates exclusively through input quantities, then expected procurement costs satisfy: $\mathbb{E}[C^{\mathcal{A}_{FP}}] > \mathbb{E}[C^{\mathcal{A}_{CS}}]$. Moreover,*

(a) if the environment is elastic, then: $\mathbb{E}[C^{\mathcal{A}_S}] > \mathbb{E}[C^{\mathcal{A}_{FP}}]$.

(b) if the environment is inelastic and $q'_t(\theta) \geq 0$ for every task t and type θ , then:

$\mathbb{E}[C^{\mathcal{A}_{FP}}] \geq \mathbb{E}[C^{\mathcal{A}_S}]$, with strict inequality if $q'_t(\cdot) > 0$ for each t .

In the heterogeneous-quantities environment of part 3(b), the ranking between the scaling and cost-sharing auctions is generally ambiguous when sufficiently low relative weight is placed on an inelastic task.¹¹ Meanwhile, the ranking in part 3(a) is an ex-ante comparison and need not hold for every realized profile of seller types.¹²

With risk aversion, the comparison across auction formats becomes more intricate. Procurement costs reflect not only allocative efficiency and the linkage effect, but also insurance and the resulting “income and substitution effects” on bidding incentives. These forces interact in ways that need not produce a simple ranking. We therefore turn to a matched set of examples that hold the distribution of total project costs fixed while varying the structure of quantity risk. Even with the same distribution of total costs, scaling auctions can span the entire cost ranking, ranging from the least to the most expensive format depending on how stochasticity enters the environment.

3.3 Risk Aversion and Correlated Quantity Shocks

The procurement cost comparisons provided in Section 3.2 assume that all sellers are risk-neutral. In this section we use a matched set of examples to illustrate our results and

¹¹Consider an environment with $I = 2$, $\theta \sim \text{Unif}[0, 1]$, $(c_1, c_2) = (150, 100)$, $k = 500$, cost-sharing transfer $X = 1200$, scaling weights $\mathbf{w} = (20, 1)$, and common shock $(\bar{q}_1, \bar{q}_2) = (6, 2)$ with probability $\frac{1}{2}$. Base input quantities $(20\theta, \theta)$ yield $\mathbb{E}[C^{\mathcal{A}_{CS}}] < \mathbb{E}[C^{\mathcal{A}_S}] < \mathbb{E}[C^{\mathcal{A}_{FP}}]$, while $(15\theta, 2\theta)$ yield $\mathbb{E}[C^{\mathcal{A}_S}] < \mathbb{E}[C^{\mathcal{A}_{CS}}] < \mathbb{E}[C^{\mathcal{A}_{FP}}]$. In both environments, task 2 is inelastic, bidders bid only on task 2, and the scaling auction is well behaved.

¹²The ex-ante ranking in part 3(a) is driven by both allocative inefficiency and informational rents. After types are realized, however, the ranking may reverse. In particular, when all realized sellers are of the most efficient type, the project is allocated efficiently under every format, while informational rents vanish only under the scaling auction. Scaling can therefore be buyer-optimal for such type profiles.

show how risk aversion can change these comparisons. In particular, we show how different joint distributions of input quantities can lead to substantially different risk allocations and thus procurement costs *even with the same conditional distributions of total costs*.

Suppose that there are two sellers with types independently and uniformly distributed on the unit interval. There are two tasks, each input realization occurs with probability one half, the weights in the scaling auction are $w = (1, 1)$, and the fixed transfer in the cost-sharing auction is $X = 12$. The quantity shock ξ^q varies between the environments, with equal chance of taking values $(0, 0)$ and $(2, 2)$ under fixed-cost and input-quantity heterogeneity, values $(0, 0)$ and $(4, 2)$ under marginal-cost heterogeneity and positive correlation, and $(0, 4)$ and $(4, 0)$ under marginal-cost heterogeneity and negative correlation. There are no fixed-cost or marginal-cost shocks ($\xi^c = (0, 0)$ and $\xi^k = 0$). Table 1 describes four environments. In all environments, total costs conditional on type θ are either $12 + \theta$ or $16 + \theta$, each with equal chance.

Table 1: Four illustrative environments

Source of heterogeneity	$k(\theta)$	$c(\theta)$	$q(\theta) + \underline{\xi}^q$	$q(\theta) + \bar{\xi}^q$
Fixed costs	$10 + \theta$	$(1, 1)$	$(1, 1)$	$(3, 3)$
Input quantities	10	$(1, 1)$	$(1 + \frac{1}{2}\theta, 1 + \frac{1}{2}\theta)$	$(3 + \frac{1}{2}\theta, 3 + \frac{1}{2}\theta)$
Marginal costs (+)	10	$(\frac{2-\theta}{3}, \frac{2+2\theta}{3})$	$(1, 2)$	$(5, 4)$
Marginal costs (−)	10	$(\frac{7+\theta}{6}, \frac{1+\theta}{6})$	$(1, 5)$	$(5, 1)$

Notes: The low and high input realizations each occur with probability one half. The symbols (+) and (−) denote positively and negatively correlated input quantities, respectively.

Risk-neutral benchmark. Because all four environments have the same total-costs, they all have the same equilibrium in the fixed-payment auction. Environments 2-4 have the same

variable cost, so they share equilibria under the cost-sharing auction.¹³

Table 2 summarizes the expected procurement costs across the environments and auction formats under risk neutrality. The table demonstrates the results identified in Theorem 2. When heterogeneity operates through fixed costs, all formats generate the same expected procurement costs. When heterogeneity operates through input quantities, the environment is elastic and the scaling auction induces perverse incentives and has the highest procurement costs. Finally, when heterogeneity operates through marginal costs, the cost-sharing auction generates strictly lower procurement costs than the fixed-payment or scaling auction.

Table 2: Expected procurement costs in the illustrative environments

Source of heterogeneity	Fixed-Payment	Cost-Sharing	Scaling
Fixed costs	$\frac{44}{3} \approx 14.67$	$\frac{44}{3} \approx 14.67$	$\frac{44}{3} \approx 14.67$
Input quantities	$\frac{44}{3} \approx 14.67$	$\frac{22}{3} + 32 \ln\left(\frac{5}{4}\right) \approx 14.47$	$250 \ln\left(\frac{5}{4}\right) - \frac{121}{3} \approx 15.45$
Marginal costs, (+)	$\frac{44}{3} \approx 14.67$	$\frac{22}{3} + 32 \ln\left(\frac{5}{4}\right) \approx 14.47$	$\frac{44}{3} \approx 14.67$
Marginal costs, (−)	$\frac{44}{3} \approx 14.67$	$\frac{22}{3} + 32 \ln\left(\frac{5}{4}\right) \approx 14.47$	$\frac{44}{3} \approx 14.67$

Correlation and residual risk. We now evaluate how risk aversion affects behavior in the two marginal-cost environments. Because both environments generate the same conditional distributions of total and variable costs, they produce identical payoffs under fixed-payment and cost-sharing. Thus, even risk-averse sellers should behave exactly the same across the two environments.

The correlation structure matters to a risk-averse bidder under scaling. Under positive correlation, changing the allocation of a given score across the two task-specific bids can affect the distribution of payoffs but cannot eliminate the risk associated with the quantity shock. This contrasts with the negative correlation environment, where even under a fixed score, a seller can completely eliminate post-award risk by choosing bids that satisfy $b_1 - b_2 = 1$.

¹³We explicitly state and verify the equilibria in Online Appendix C.

As a result, a risk-averse seller will tend to bid more aggressively in the negative-correlation environment.

Figure 1 shows how the buyer's procurement costs vary with the level of risk aversion when sellers have CARA utility. Several features stand out. First, as clear from Table 2, cost-sharing auctions deliver lower procurement costs under risk neutrality than fixed-payment and scaling auctions under either correlation environment. This is the result of the linkage principle under risk neutrality. Procurement costs *fall* monotonically with risk aversion in the negative-correlation scaling environment, while they generally rise with risk aversion in the other environments over the range considered.¹⁴ For high enough values of α , expected procurement costs under scaling auctions in the negative correlation environment are actually lower than those in the cost-sharing auction.

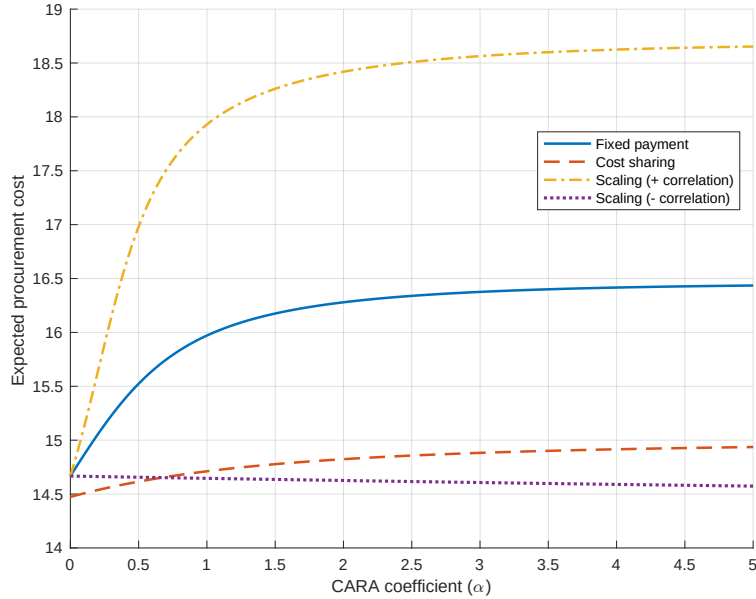


Figure 1: Procurement costs in marginal costs environments as a function of CARA parameter

¹⁴The intuition behind this pattern is similar to that of risk aversion in the classic first-price auction setting, where more risk averse buyers bid more aggressively because they are more willing to give up payoffs in the winning state in order to increase the likelihood of winning.

4 Experimental design

We designed and carried out two laboratory experiments to test the main predictions of the theoretical results discussed in Section 3. In our first experiment, we compared cost-sharing and scaling auctions in an environment in which heterogeneity operated exclusively through input quantities. In our second experiment, we compared fixed-payment, cost-sharing, and scaling auctions in an environment in which heterogeneity operated exclusively through marginal costs. Experiment 1 was carried out in October 2025 and Experiment 2 in March 2026, both in the University of Queensland’s Centre for Unified Behavioural and Economic Sciences Laboratory. Both experiments were preregistered in the American Economic Association’s RCT Registry (AEARCTR-0017041 and AEARCTR-0017929).

4.1 Treatments

The experiments were framed in a procurement setting. Subjects were told that a government was holding an auction to determine who would complete a project and how much the contractor would be paid. The project required two inputs, the realized quantities of which were uncertain and had to be provided by the auction winner.

Experiment 1 involved a between-subject design, in which subjects were assigned either to the scaling-auction treatment or to the cost-sharing-auction treatment. There were no differences in the economic environment other than the auction’s rules. The environment was designed so that bidder heterogeneity operated through input quantities rather than marginal costs: each bidder had a privately known personal add-on that increased the quantities of both required inputs. Subjects received new types in each round. The numerical parameterization of the environment and auction rules is summarized in Column E1 of Ta-

ble 3.¹⁵

Experiment 2 was a mixed between- and within-subject design. The between-subject component crossed two dimensions: the non-fixed-payment auction format and the correlation structure of the input shocks. The auction-format dimension assigned subjects either to the scaling-auction treatment or to the cost-sharing-auction treatment. The correlation dimension assigned subjects either to a positive-correlation environment, in which the two input shocks moved together, or to a negative-correlation environment, in which a high realization of one input was paired with a low realization of the other. The within-subject variation came from the auction format implemented in each round: odd-numbered rounds were always fixed-payment auctions, while even-numbered rounds were either cost-sharing or scaling auctions, depending on the subject’s between-subject treatment.¹⁶ The numerical parameterization of the two correlation environments and auction rules is summarized in Columns E2(+) and E2(−) of Table 3.

4.2 Implementation

In this section, we discuss the implementation of both experiments. Because the experimental designs shared many characteristics, we discuss them jointly.

The sequence of activities in both experiments was the same. Subjects were brought into a computer lab and asked to review and sign consent forms. An experimenter then read

¹⁵The richness condition is sufficient (but not necessary) for the solution of the differential equation (4) (Proposition 1) to be in the auction’s set of available scores. While the cost-sharing auction is not rich in the environment of Experiment 1, the solution of Equation (4) lies in the set of available scores. Consequently, expected costs can be analyzed using Theorem 2.

¹⁶Our experimental design did not counterbalance auction formats across odd- and even-numbered rounds. As a result, comparisons between fixed-payment and contingent-payment auctions rely on the maintained assumption that there are no systematic effects associated with their position in the alternating round sequence. Our preregistered statistical test concerns the differential effect of correlation structure across the contingent payment formats and thus is not subject to this concern.

Table 3: Mapping the theoretical environment to the experimental parameterizations

Primitive / notation	E1	E2(+)	E2(-)
Number of bidders, I	2	2	2
Number of tasks, T	2	2	2
Fixed cost, $k(\theta)$	500	2400	2400
Base input quantity $\mathbf{q}(\theta)$	$(10\theta, 10\theta)$	$(0, 0)$	$(0, 0)$
Low Shock $\underline{\xi}$	$(0, 0)$	$(2, 2)$	$(1, 5)$
High Shock $\bar{\xi}$	$(6, 2)$	$(6, 6)$	$(5, 1)$
Marginal cost, task 1	150	$320 + 80\theta$	$560 + 140\theta$
Marginal cost, task 2	100	$160 + 40\theta$	$80 + 20\theta$
Scaling-auction weights $\mathbf{w}$	$(8, 6)$	$(1, 1)$	$(1, 1)$
Cost-sharing fixed transfer X	1800	4400	4400

Notes: Columns **E1**, **E2(+)**, and **E2(-)** denote Experiment 1, Experiment 2 with positively correlated shocks, and Experiment 2 with negatively correlated shocks. In both experiments, the type θ was independently and uniformly distributed on the unit interval. In each experiment, $\underline{\xi}$ and $\bar{\xi}$ each occur with probability $\frac{1}{2}$. In both experiments, the realized project quantities are given by $\mathbf{Q} = \mathbf{q}(\theta) + \xi$.

the instructions out loud as subjects viewed them on their screens and in hard copy. After the instructions, subjects were invited to review two examples.¹⁷ Subjects then completed a five-question comprehension quiz, for which they were paid \$1 for each correct answer, and reviewed the solutions. They then participated in 30 rounds of auctions corresponding to their experiment and treatment. Auction outcomes were denominated in points, and one round was randomly selected for payment at the end of the experiment. Points from the selected round were converted to dollars at a rate of 150 points per dollar in Experiment 1 and 200 points per dollar in Experiment 2. After being informed of their payment, subjects completed a demographic survey, a cognitive reflection test (Frederick, 2005), and a short survey about the experiment.

In each round, subjects were randomly assigned a new partner. On each auction page, subjects were informed of their random type and reminded of how payoffs would be calculated. Subjects made bids using sliders.¹⁸ To make the payoff consequences of each bid transparent, the interface updated dynamically as subjects moved the sliders, showing the likelihood of each input shock and the payoffs conditional on winning under each possible input realization. After choosing their bid with the slider, subjects submitted their bid by clicking next. Figure 2 illustrates the scaling-auction interface, including the bid sliders, score calculation, and payoff information shown to subjects.

¹⁷In Experiment 1, both examples related to the single auction format relevant for the subject’s treatment. In Experiment 2, subjects saw one example of a fixed-payment auction and one example of either a cost-sharing or unit-price (scaling) auction, whichever was relevant for their treatment.

¹⁸One implementation issue is that each auction format requires a choice of admissible bid range. We chose ranges that contained all risk-neutral Nash equilibrium bids and allowed substantial deviations beyond them. In Experiment 1, subjects could bid up to 600 per task in scaling auctions and up to 100% in cost-sharing auctions. In Experiment 2, maximum bids were 10,000 in fixed-payment auctions, 1,600 per task in scaling auctions, and 50% in cost-sharing auctions. All minimum bids were zero.

Auction Round 1

Your personal input add-on for this round is 1.2. That means that

- If the baseline requirement is low, you will require 1.2 units of Input 1 and 1.2 units of Input 2.
- If the baseline requirement is high, you will require 7.2 units of Input 1 and 3.2 of Input 2.

Set your **bid per unit** for Input 1 and Input 2 using the sliders. These bids will be combined into a score using fixed weights. The **lowest score wins**.

If you win: $7,500 + (\text{your bid} - \text{cost}) \times \text{units} - 500$. Variable costs are 150 per unit of Input 1 and 100 per unit of Input 2.

If you lose: you keep 7,500.

You see only your own input quantities. Your opponent's are hidden.

0		600
Input 1 Bid: 220		
0		600
Input 2 Bid: 300		

Your score for this auction is $8 \times 220 + 6 \times 300 = 3560$. Remember that the bidder with the lowest score wins.

Click "Next" when you're ready to submit your bid.

Next

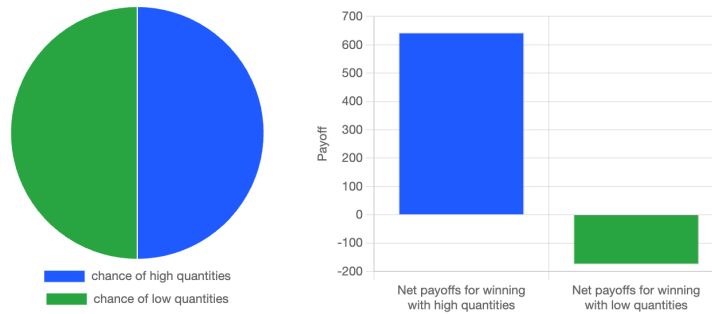


Figure 2: Bidding Page in the scaling treatment of Experiment 1

After both bids were submitted, subjects saw the outcome of the auction. They were informed of their own bid, their opponent's bid, and whether they had won. If the subject lost the auction, they were informed of their payoff. If the subject won the auction, they were informed of their payoff under each possible input-shock realization and were reminded of the payoff of the losing bidder. Thus, before the payment round was selected, winning subjects observed the payoff consequences of each possible realization rather than a single realized project cost.

A total of 102 subjects participated in Experiment 1 and 146 subjects in Experiment 2. Experiment 1 consisted of six sessions: three in the scaling treatment and three in the cost-sharing treatment. Experiment 2 consisted of eight sessions, with 2 sessions in each between-subject treatment cell. Summary statistics for subjects' demographic characteristics can be found in Appendix Table 6. All sessions for both experiments were booked for two hours. Including the completion fees of \$20, average earnings were \$56.05 for Experiment 1 and \$53.80 for Experiment 2.

4.3 Hypotheses

Here, we present the main hypotheses our experiments are designed to test.

Hypothesis 1 (Experiment 1). *When heterogeneity operates exclusively through input quantities, cost-sharing auctions generate higher efficiency and lower procurement costs than scaling auctions.*

Hypothesis 1 concerns the outcomes of the auctions in Experiment 1. According to the definition from Section 3.1, both tasks in the economic environment used in this experiment are elastic. Thus, according to Theorem 1, in cost-sharing auctions the more efficient seller is expected to win, while in scaling auctions the less efficient seller is expected to win. Moreover, Part 3 of Theorem 2 implies that expected procurement costs should be lower in cost-sharing auctions.

Hypothesis 2a (Experiment 2: Risk-neutral benchmark). *When heterogeneity operates exclusively through marginal costs and bidders are risk neutral, all auction formats are efficient. Cost-sharing auctions generate lower procurement costs than scaling and fixed-payment auctions, which generate equivalent costs.*

Hypothesis 2a states the implications of our theoretical results for efficiency and costs in Experiment 2. It is the direct implication of Part 2 of Theorem 1 and Part 2 of Theorem 2. Note that in contrast to the predictions for Experiment 1, we do not expect differences in efficiency between the auction formats in Experiment 2 under risk neutrality.

Hypothesis 2b (Experiment 2: Risk-aversion alternative). *When heterogeneity operates exclusively through marginal costs and bidders are sufficiently risk averse, fixed-payment and cost-sharing auctions are efficient. Expected procurement costs are highest in scaling auctions with positively correlated inputs, followed by fixed-payment auctions, cost-sharing auctions, and scaling auctions with negatively correlated inputs.*

Hypothesis 2b provides an alternative to Hypothesis 2a that accounts for the potential for risk aversion. Even with risk aversion we expect fixed-payment and cost-sharing auctions to be efficient, while we no longer have a sharp prediction for scaling auctions. However, as described in the examples in Section 3.3, risk aversion is expected to interact with the task correlation structure to affect expected procurement costs. In particular, scaling auctions are expected to amplify the inherent riskiness of the positive correlation treatment, leading risk-averse sellers to bid less aggressively and increase procurement costs. In contrast, scaling auctions allow sellers to *completely* eliminate post-award risk in the negative correlation treatment, leading us to expect more aggressive bidding and low procurement costs. Because cost-sharing auctions provide some insurance to risk-averse sellers, they are expected to lead to lower procurement costs. Notably, the treatments are constructed such that the post-award distribution of payoffs conditional on bids does not differ between the positive and negative correlation cases, so we do not expect this treatment dimension to affect bidding, efficiency, or procurement costs for cash or cost-sharing auctions.

5 Results

We organize the results by experiment. For each experiment we examine allocative efficiency, government procurement costs, and bidding behavior. We preregistered the regressions reported in Table 4 and columns (2) and (4) of Table 5. In particular, our pre-analysis plans commit to use the data from rounds 6-30 in Experiment 1 and 5-30 in Experiment 2. All other analysis should be viewed as exploratory.

Because some treatments were assigned at the session level, we also report p-values from two-sided randomization-inference (RI) tests. For Tables 4 and 5, we re-estimate each regression under all possible assignments of the between-subject treatments (20 and 2520 assignments, respectively). For each coefficient or reported contrast, the RI p-value is the fraction of these assignments yielding an absolute session-clustered t statistic at least as large as the absolute value of the observed statistic. Thus, each of these is a p-value for the sharp null that treatment assignment has no effect on the outcome under study.

5.1 Experiment 1: Types as Quantities

5.1.1 Efficiency and Government Costs

We estimate the following equation to assess allocative efficiency and procurement costs:

$$y_{jt} = \beta_0 + \beta_1 \text{Scaling}_j + \beta_2 \text{High Type}_{jt} + \beta_3 \text{Low Type}_{jt} + \gamma_t + \varepsilon_{jt}. \quad (2)$$

Index j refers to the group and index t to the round. The outcome variable y_{jt} refers to one of two auction outcomes: (1) the post-award expected costs of the winner, or (2) the

post-award expected procurement costs.¹⁹ Scaling_j is an indicator variable equal to one if group j is in the Scaling treatment. High Type_{jt} and Low Type_{jt} are the inefficient and efficient (respectively) types in group j and round t . The coefficient γ_t captures round fixed effects.

Table 4: Experiment 1: Efficiency and Procurement Costs

	(1) Total Costs	(2) Procurement Costs
Scaling	534.3*** (32.9) [0.100]	638.4*** (41.7) [0.100]
Controls	Yes	Yes
Round Fixed Effects	Yes	Yes
R-Sqd	0.64	0.45
Observations	1275	1275

Notes: Standard errors clustered at the session level are in parentheses; two-sided randomization-inference p -values are in square brackets. Significance stars are based on the session-clustered t -tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The results are reported in Table 4.²⁰ Column 1 shows that costs of the winning type are approximately 534 points higher under scaling than under cost-sharing. Since sellers' expected costs are drawn uniformly on $[1050, 3550]$, a fully efficient auction would yield an expected cost of 1883.33, while a fully inefficient auction—one that always selects the highest-cost bidder—would yield expected costs of 2716.67. The estimated coefficient therefore implies that switching from cost-sharing to scaling moves the allocation roughly two-thirds

¹⁹Procurement costs in scaling auctions are calculated by multiplying the bids of the winner by the winner's expected quantities. Procurement costs in cost-sharing auctions are calculated as the up-front payment plus the product of the winner's bid and expected variable costs.

²⁰With three sessions per treatment, there are $\binom{6}{3} = 20$ possible assignments. Swapping all treatment labels changes the sign of the treatment t statistic but not its absolute value, so the smallest attainable two-sided RI p -value is $2/20 = 0.10$. The RI p -values for the treatment variable attain this minimum in both columns.

of the way from maximum efficiency to maximum inefficiency, a quantitatively large effect consistent with the perverse-incentives prediction in Hypothesis 1.

Column 2 presents the effect of auction design on procurement costs. The coefficient on scaling indicates that the government pays substantially more under the scaling treatment relative to cost-sharing. Theory predicts that moving from cost-sharing to scaling increases expected procurement costs from 2,355.11 to 3,081.87, an increase of 726.76 points. The empirical estimate is close to this prediction, providing direct experimental support for the cost-ranking in Hypothesis 1.

Figure 3 provides complementary evidence on efficiency by showing the proportion of auctions in which the more efficient type wins as a function of the difference between the two types in a given group.

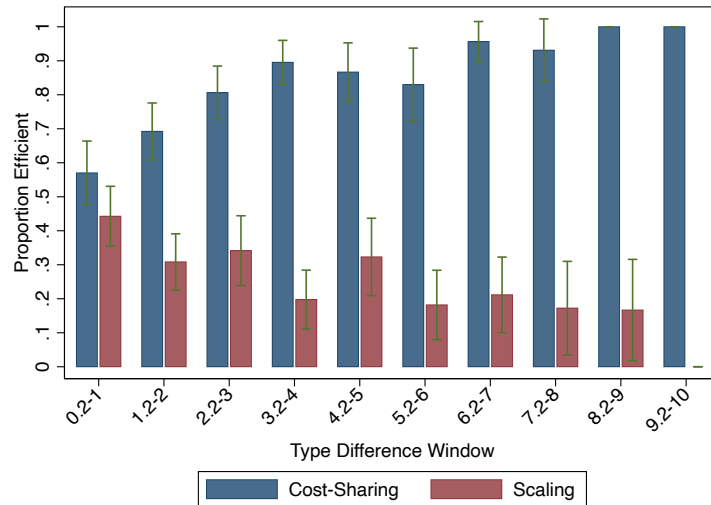


Figure 3: Proportion of auctions in which the efficient type wins, by type gap and auction format (Experiment 1). Data from auctions in which both bidders had the same type are omitted. Theory predicts all blue bars at 1 and all red bars at 0.

When types are close, both auction formats award the contract to the more efficient

bidder with close to 50% probability, as one would expect in a competitive auction with little differentiation. As the gap between types widens, the two mechanisms diverge sharply. Under cost-sharing, the probability that the more efficient type wins increases toward 100%, consistent with the well-behaved equilibrium prediction. Under Scaling, the probability moves in the opposite direction—toward 0%—consistent with the perverse equilibrium in which less efficient types bid more aggressively. Theory predicts the cost-sharing bars to all equal one and the Scaling bars to all equal zero; the data conform closely to these directional predictions.

Result 1. *In Experiment 1, cost-sharing auctions generated higher efficiency and lower procurement costs than scaling auctions. Thus, our evidence supports Hypothesis 1.*

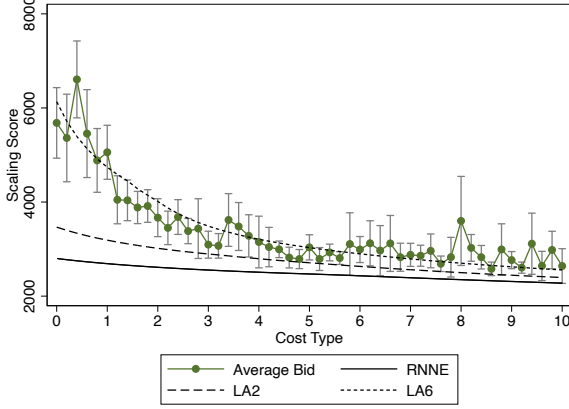
5.1.2 Bidding Behavior

Figure 4 displays average bids in the cost-sharing auction and average scores in the Scaling auction, plotted against bidder type, alongside the theoretical equilibrium predictions (shown using lines without markers). These average bidding functions support the key mechanism behind Hypothesis 1: average scores are decreasing in type in scaling auctions, and average bids are increasing in type in cost-sharing auctions.

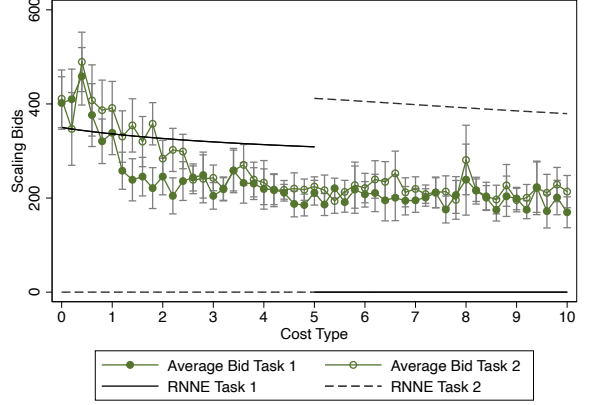
Bids, especially in scaling auctions, are higher than those that would be predicted by the risk-neutral Nash Equilibrium (RNNE). Because these auctions expose sellers to post-award risk, the theoretical predictions shown in Figure 4 also display equilibrium predictions under the assumption that sellers bid with loss-averse utility functions.²¹

²¹Specifically, these are the bidding functions when sellers have the Bernoulli utility function

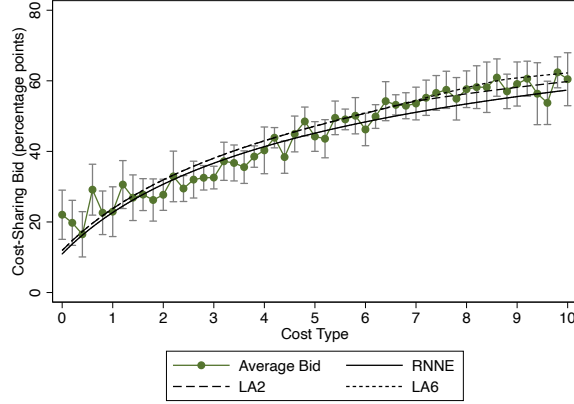
$$U(z|r) = \begin{cases} \lambda(z - r) & \text{if } z \leq r \\ z - r & \text{otherwise} \end{cases}$$



(a) Scaling Score



(b) Scaling Bids



(c) Cost-sharing Bids

Figure 4: Average Bidding Behavior in Experiment 1. Average bidding for each cost type is plotted in green with a 95% confidence interval. The RNNE bid is presented in solid black while equilibrium bidding functions under loss aversion parameters of $\lambda = 2$ and $\lambda = 6$ are presented with dashed lines.

Average cost-sharing bids are well explained by conventional levels of loss aversion, whereas matching average scaling scores requires a loss-aversion parameter of roughly 6,

for loss-aversion parameter $\lambda \geq 1$ and reference point r equal to the payoffs sellers receive for losing the auction. The reference point is fixed at the payoff from losing the auction to obtain a simple bidding benchmark. This differs from models of expectations-based loss aversion in auctions, in which the reference point is determined by the bidder's expectations (see, e.g., [Balzer and Rosato \(2021, 2025\)](#)). Our choice is motivated by Appendix Figure 6, which shows that the lower of sellers' two possible winning payoffs is often very close to the outside option, a pattern which is consistent with subjects avoiding any losses

which is high relative to most experimental estimates (Brown et al., 2024). One possible explanation is that bidders do not fully optimize the composition of their per-task bids. Average per-task bids are shown in Panel (b) of Figure 4. For low-cost types, a one-point increase in the bid for task 1 raises expected payoffs by up to three times as much as a one-point increase in the bid for task 2, while increasing the score by only 33% more. For high-cost types, the analogous trade-off favors task 2 over task 1. Thus, payoff maximization at a given score typically requires a substantially asymmetric bid vector.²² In the data, however, bids for the two tasks are relatively similar. These compositional errors are not especially costly conditional on winning: reallocating each observed score to the payoff-maximizing corner solution would increase winner expected payoffs by less than one dollar per round on average. Nevertheless, because the observed bid vectors deliver lower expected payoffs for any given score, this misallocation can help explain why subjects submit less aggressive scaling scores.

5.2 Experiment 2: Types as Marginal Costs

5.2.1 Efficiency and Government Costs

Using the data from Experiment 2, we estimate the following equation to assess the effects of auction design and the interaction of auction design and economic environment on auction outcomes:

$$y_{sjt} = \beta_1 S_{st} + \beta_2 CS_{st} + \beta_3 (S_{st} \times \text{Neg}_s) + \beta_4 (CS_{st} \times \text{Neg}_s) + \gamma_s + \theta' Z_{sjt} + \varepsilon_{sjt}. \quad (3)$$

²²Appendix Figure 7 shows per-task bidding functions under loss aversion; these predictions are not substantially closer to average bidding than the risk-neutral prediction.

Index s refers to the session, index j to the group, and index t to the round. The outcome variable y_{sjt} refers to one of three auction outcomes: (1) the post-award expected costs of the winning bidder, (2) the post-award expected costs to the government, or (3) the absolute difference between the winner’s two possible payoffs.²³ S_{st} and CS_{st} are indicators for session s in round t being either a scaling or cost-sharing auction, respectively. Neg_s is an indicator for session s being a session in the negative correlation environment. γ_s refers to session fixed effects. Z_{sjt} is a vector containing the highest and lowest cost types in the auction as well as the round number. The results of estimating this equation can be found in Table 5.

Columns 1 and 2 address allocative efficiency. The estimates indicate that in this setting, cost-sharing auctions are slightly more efficient than fixed-payment, and fixed-payment slightly more efficient than scaling. We do not find significant interaction effects with the correlation structure. While the results are statistically significant, these efficiency differences are quantitatively modest: bidder costs are distributed uniformly over the interval $[4320, 4800]$, so a shift of 15 units corresponds to moving approximately 10% of the way from maximum efficiency to maximum inefficiency.

Columns 3 and 4 address procurement costs. Column 3 shows that without correlation interactions, fixed-payment and scaling auctions generate similar costs for the government, while cost-sharing auctions are significantly more expensive. However, this overall comparison masks important heterogeneity across correlation environments. Column 4 reveals a significant interaction: under negative correlation between tasks, the scaling auction performs substantially better than fixed-payment, reducing government costs; under positive correlation, scaling performs worse than fixed-payment. This pattern is consistent with the

²³In fixed-payment auctions, the cost to the government is simply the winning bid. In Scaling auctions, the cost to the government is calculated based on multiplying the bids of the winner by the winner’s expected quantities. In cost-sharing auctions, it is the up-front payment plus the product of the winner’s bid and variable costs.

Table 5: Experiment 2 Outcomes

	(1) TC	(2) TC	(3) PC	(4) PC	(5) PD	(6) PD
Scaling	15.0** (5.21) [0.057]	18.4 (10.4) [0.436]	3.51 (383.4) [0.986]	736.5*** (150.2) [0.262]	53.9 (608.9) [0.843]	1250.4** (419.0) [0.367]
Cost-Sharing	-15.3* (7.18) [0.086]	-24.7*** (1.11) [0.038]	942.4*** (250.3) [0.143]	1040.4* (473.7) [0.562]	-338.2*** (33.9) [0.014]	-326.0*** (63.8) [0.233]
Scaling $\times$ Neg		-6.33 (10.3) [0.610]		-1355.8*** (257.0) [0.041]		-2213.3*** (421.4) [0.030]
Cost-Sharing $\times$ Neg		17.8 (10.0) [0.297]		-185.8 (507.4) [0.835]		-23.2 (68.7) [0.863]
t-test p-value: $S \times N = C \times N$		0.13		0.079		0.0014
RI p-value: $S \times N = C \times N$		0.21		0.22		0.0095
Controls	Yes	Yes	Yes	Yes	Yes	Yes
R-Sqd	0.52	0.52	0.18	0.21	0.22	0.39
Observations	1898	1898	1898	1898	1898	1898

Notes: “TC” stands for “total costs,” “PC” stands for “procurement costs,” and “PD” stands for “payoff difference.” Standard errors clustered at the session level are in parentheses; two-sided randomization-inference p -values are in square brackets. Significance stars are based on the session-clustered t -tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

theoretical prediction that the insurance value of scaling auctions is highest when project risks are negatively correlated across tasks. The prespecified difference between the scaling and cost-sharing interaction effects is in the predicted direction and significant at the 10% level with session-clustered inference ($p = 0.079$), but not with randomization inference ($p = 0.22$).²⁴

²⁴Appendix Table 9 reports the results of regressing the auction outcome variables on treatments using the data from only contingent auctions (and thus not using the Fixed-Payment auctions as a baseline). For procurement costs, the difference in interaction effects is similar: -1289 points, as compared to the -1170 points in Table 5. In this supplementary regression, the difference is significant at the 5% level using both session-clustered and randomization inference.

Result 2. *In Experiment 2, differences in allocative efficiency across treatments are small. Procurement costs in scaling auctions are lower under negative correlation than positive correlation, a result that is inconsistent with the risk-neutral cost prediction in Hypothesis 2a and consistent with the risk-sharing mechanism in Hypothesis 2b. Cost-sharing does not reduce procurement costs relative to fixed-payment auctions.*

The large coefficient on Cost-Sharing from Column 3 warrants further explanation. A winner in Experiment 2's cost-sharing auctions receives a fixed transfer of 4400 points in addition to a share of variable costs. Thus, in contrast to fixed-payment and scaling auctions, 4400 is a floor on procurement costs for even the most aggressive bids in cost-sharing auctions. Two empirical exercises help interpret the estimated differences. First, Appendix Table 9 restricts the regression to contingent auctions, thus removing the fixed-payment benchmark. The pooled procurement costs of scaling auctions are estimated to be 99 points higher than cost-sharing auctions, and this difference is not statistically significant. In a second exercise reported in Appendix Table 10, we replace each bid with the average bid conditional on cost type, auction format, and correlation treatment, then recompute auction outcomes. The resulting average procurement costs are 4836 for cost-sharing auctions and 4873 for fixed-payment auctions. These results suggest that low bids in the fixed-payment auctions contributed to the large difference in Column 3.

Columns 5 and 6 of Table 5 address winners' post-award risk exposure. Higher values of the absolute difference between the two possible payoff outcomes indicate that the auction design is exposing sellers to more risk. Column 5 shows that scaling and fixed-payment auctions leave winners with similar residual risk, while cost-sharing auctions significantly reduce winners' post-award payoff differences. The interaction terms reveal a pattern that closely aligns with the mechanism behind Hypothesis 2b: Scaling auctions are highly effective

at providing insurance under negative correlation—where payment and cost fluctuations partially offset each other—but actually increase risk exposure under positive correlation, where payments scale with quantities more than costs do, widening the gap between the two payoffs. Cost-sharing auctions reduce risk uniformly across both correlation conditions. We explore how this risk affects bidding behavior in the next subsection.

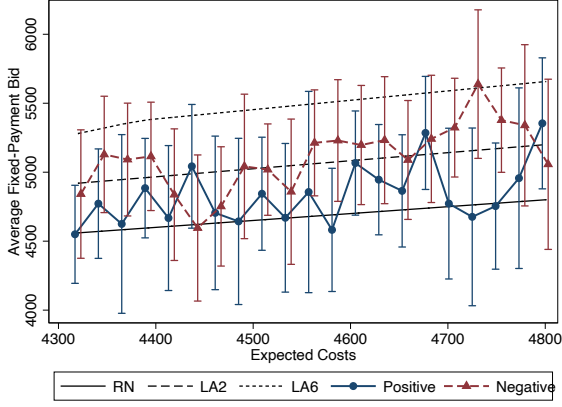
5.2.2 Bidding Behavior

Figure 5 plots average bidding functions for each auction format, separately for the positive and negative correlation conditions.²⁵ The lines without markers provide the theoretical predictions for bidding under risk neutrality and loss aversion with parameters 2 and 6.

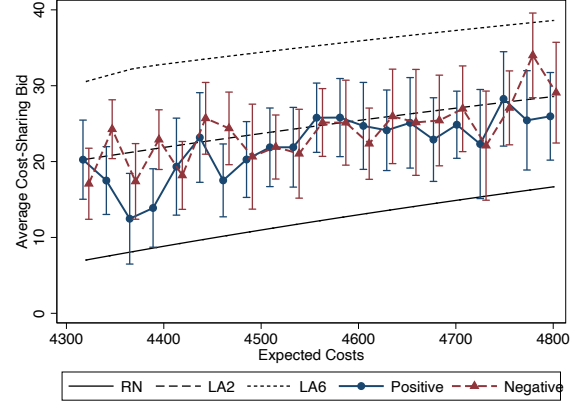
Figure 5 suggests that departures from risk-neutral bidding are closely related to residual payoff risk. In fixed-payment and cost-sharing auctions, average bids in both correlation treatments are generally close to the predictions generated by loss aversion with parameter 2. Scaling auctions show sharper variation. Under positive correlation, scaling leaves winners exposed to substantial residual payoff risk, and average scores are closer to the more conservative benchmark generated by loss aversion with parameter 6.²⁶ Under negative correlation, scaling allows bidders to hedge this risk through the composition of their unit-price bids. Loss aversion predicts the same score as risk neutrality, but also predicts substantially higher bids for task 1 than task 2. The data move in this direction, although the observed gap is smaller than the theoretical hedge. Thus, the scaling results are consistent with the same risk-based mechanism: residual risk raises scores under positive correlation, while the

²⁵Empirical average scores alongside the theoretical predictions can be found in Appendix Figure 8.

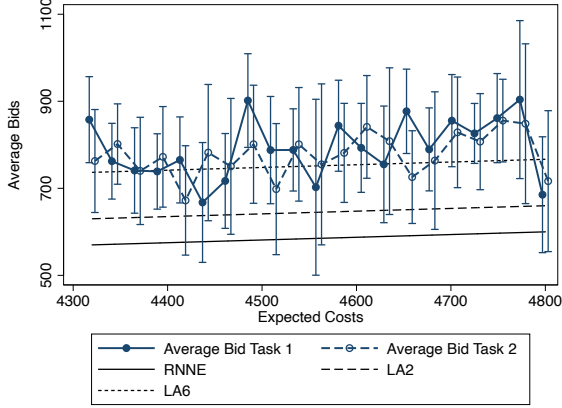
²⁶Because the two quantities always move together in the positive-correlation treatment and the score is the unweighted sum of task bids, only the sum of the two unit-price bids is payoff-relevant. All bid vectors with the equilibrium sum are therefore equivalent; an equal split is a natural focal allocation, and this is approximately what subjects choose on average.



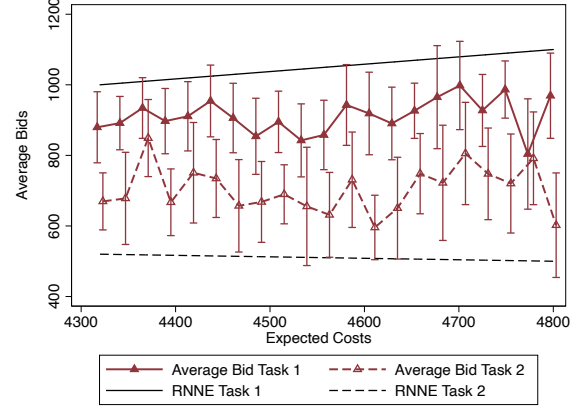
(a) Fixed-payment Bid (Both Correlations)



(b) Cost-sharing Bid (Both Correlations)



(c) Scaling Bids (Positive Correlation)



(d) Scaling Bids (Negative Correlation)

Figure 5: Average Bidding Behavior in Experiment 2. Average bidding under positive correlation is plotted using blue circles and under negative correlation red triangles. The RNNE is presented in solid black while equilibrium bidding functions under loss aversion parameters of $\lambda = 2$ and $\lambda = 6$ are presented with dashed lines. In panel (c), we assume that subjects split their score equally between the two tasks. In panel (d), equilibrium under loss aversion is the same as the RNNE.

opportunity to hedge induces asymmetric, but incomplete, bid adjustment under negative correlation.

6 Concluding remarks

Procurement mechanisms do more than determine how much the buyer pays: they shape which firms win, how project risk is shared, and how closely payments track realized costs. This paper develops a unified framework for studying these forces in contingent procurement auctions and uses it to compare fixed-payment, scaling, and cost-sharing mechanisms. The central message is that the performance of a procurement format cannot be evaluated independently of the underlying source of firm heterogeneity. The same contingent-payment rule can improve risk sharing in one environment and generate severe adverse selection in another.

Our theoretical and experimental results make this trade-off transparent. Scaling auctions can provide powerful insurance and, when sellers can use multidimensional bids to hedge project risk, can outperform conventional fixed payments. Yet scaling is also uniquely vulnerable to a perverse form of selection in which less efficient firms bid more aggressively and are more likely to win. Cost sharing provides a useful alternative: it preserves the efficient ordering of sellers while combining insurance with a direct linkage between payments and realized costs. The experiments broadly validate these mechanisms, while also showing that the behavioral response to risk can materially alter the procurement-cost rankings predicted by the risk-neutral benchmark.

More broadly, our results point to a fundamental identification problem in procurement design. Whether firms differ primarily in fixed costs, marginal costs, input requirements, or exposure to risk is rarely directly observed, yet these distinctions can determine which auction format performs best. Our analysis provides a first step toward organizing this problem theoretically and experimentally. A natural next step is to identify these dimensions of heterogeneity in field data and to use them to guide the design of contingent contracts

that jointly manage selection, risk sharing, and informational rents.

References

- Albano, G. L., A. Cipollone, R. Di Paolo, G. Ponti, and M. Sparro (2024). Scoring rules in experimental procurement. *Journal of Behavioral and Experimental Economics* 108, 102131.
- Asker, J. and E. Cantillon (2008). Properties of scoring auctions. *The RAND Journal of Economics* 39(1), 69–85.
- Asker, J. and E. Cantillon (2010). Procurement when price and quality matter. *The RAND Journal of Economics* 41(1), 1–34.
- Awaya, Y., N. Fujiwara, and M. Szabo (2025). Quality and price in scoring auctions. *Journal of Mathematical Economics* 116, 103083.
- Bajoori, E., R. Peeters, and L. Wolk (2024). Security auctions with cash- and equity-bids: An experimental study. *European Economic Review* 163, 104702.
- Balzer, B. and A. Rosato (2021). Expectations-based loss aversion in auctions with interdependent values: Extensive vs. intensive risk. *Management Science* 67(2), 1056–1074.
- Balzer, B. and A. Rosato (2025). Never say never: Optimal exclusion and reserve prices with expectations-based loss-averse buyers. *Journal of Economic Theory* 228, 106045.
- Bergemann, D., T. Heumann, and S. Morris (2025, December). Procurement without priors: A simple mechanism and its notable performance. Cowles Foundation Discussion Paper 2479, Cowles Foundation for Research in Economics, Yale University.
- Bichler, M. (2000). An experimental analysis of multi-attribute auctions. *Decision Support Systems* 29(3), 249–268.

- Bolotnyy, V. and S. Vasserman (2023). Scaling auctions as insurance: A case study in infrastructure procurement. *Econometrica* 91(4), 1205–1259.
- Bos, O., N. Fugger, and S. Onderstal (2025). Profit-share auctions in procurement. Technical report, ZEW Discussion Papers.
- Breig, Z., A. Hernández-Chanto, and E. Hiller (2025). Competition between sellers in security-bid auctions: An experimental study. *Experimental Economics* 28(4), 910–935.
- Breig, Z., A. Hernández-Chanto, and D. Hunt (2026). Experimental auctions with securities. *The Economic Journal*. Forthcoming.
- Brown, A. L., T. Imai, F. M. Vieider, and C. F. Camerer (2024). Meta-analysis of empirical estimates of loss aversion. *Journal of Economic Literature* 62(2), 485–516.
- Camboni, R., L. Corazzini, S. Galavotti, and P. Valbonesi (2025). Bidding on price and quality: An experiment on the complexity of scoring rule auctions. *Review of Economics and Statistics* 107(3), 755–770.
- Che, Y.-K. (1993). Design competition through multidimensional auctions. *The RAND Journal of Economics* 24(4), 668–680.
- Chen-Ritzo, C.-H., T. P. Harrison, A. M. Kwasnica, and G. J. Thomas (2005). Better, faster, cheaper: An experimental analysis of a multiattribute reverse auction mechanism with restricted information feedback. *Management Science* 51(12), 1753–1762.
- Ewerhart, C. and K. Fieseler (2003). Procurement auctions and unit-price contracts. *The RAND Journal of Economics* 34(3), 569–581.
- Fok, C. I. (2023). *Three Essays in Industrial Organization*. PhD thesis, University of Toronto (Canada).
- Frederick, S. (2005). Cognitive reflection and decision making. *Journal of Economic Perspectives* 19(4), 25–42.

- Hanazono, M., J. Nakabayashi, R. Sano, and M. Tsuruoka (2024). Beyond quasilinearity: Exploring nonlinear scoring rules in procurement auctions. Working paper, University of Tokyo Market Design Center.
- Hernández-Chanto, A. and E. Rivera Mora (2025). Optimal scaling auctions: A consumer theory deconstruction. *Working paper*.
- Luo, Y. and H. Takahashi (2025). Bidding for contracts under uncertain demand: Skewed bidding and risk sharing. *The RAND Journal of Economics* 56(3), 325–343.
- Shachat, J., M. J. Walker, and L. Wei (2024). Contingent payments in procurement interactions: Experimental evidence. *European Economic Review* 170, 104886.
- Strecker, S. (2010). Information revelation in multiattribute English auctions: A laboratory study. *Decision Support Systems* 49(3), 272–280.
- Topkis, D. M. (1978). Minimizing a submodular function on a lattice. *Operations Research* 26(2), 305–321.

A Additional theoretical results and proofs

Bidding equilibrium derivation

Proposition 1. *Fix a rich auction $\mathcal{A}$. Suppose also that $U(\cdot, \theta \mid \mathcal{A})$ is weakly concave on S for every $\theta \in \Theta$. If $\mathcal{A}$ is well behaved for environment $\mathcal{E}$, then $\mathcal{A}$ has a unique symmetric equilibrium. This equilibrium is increasing, differentiable, and is characterized as the unique solution to the differential equation*

$$\varphi'(\theta) = \frac{(I-1)f(\theta)}{1-F(\theta)} \frac{U(\varphi(\theta), \theta \mid \mathcal{A})}{U_1(\varphi(\theta), \theta \mid \mathcal{A})} > 0, \quad (4)$$

subject to the boundary condition $U(\varphi(\bar{\theta}), \bar{\theta} \mid \mathcal{A}) = 0$. If $\mathcal{A}$ induces perverse incentives for environment $\mathcal{E}$, then $\mathcal{A}$ has a unique symmetric equilibrium. This equilibrium is decreasing, differentiable, and is characterized as the unique solution to the differential equation

$$\varphi'(\theta) = -\frac{(I-1)f(\theta)}{F(\theta)} \frac{U(\varphi(\theta), \theta \mid \mathcal{A})}{U_1(\varphi(\theta), \theta \mid \mathcal{A})} < 0, \quad (5)$$

subject to the boundary condition $U(\varphi(\underline{\theta}), \underline{\theta} \mid \mathcal{A}) = 0$.

Proof of Proposition 1

We begin by establishing monotonicity of the equilibrium bidding strategy using Topkis's Monotonicity Theorem (Topkis, 1978). Consider a seller of type θ facing an arbitrary strategy profile $\varphi : [\underline{\theta}, \bar{\theta}] \rightarrow \mathbb{R}$ of the other $I-1$ opponents, and let $P(s)$ denote the induced probability that a bid s wins against the opponents' bidding strategies. Since the opponents' types are independent of θ , $P(s)$ does not depend on type. The expected utility from bidding s is therefore $V(s, \theta) = P(s)U(s, \theta \mid \mathcal{A})$.

Log-supermodular case. By assumption, $U(s, \theta \mid \mathcal{A})$ is increasing in s , differentiable, and strictly log-supermodular on Π_+^A . Since $\log V(s, \theta) = \log P(s) + \log U(s, \theta \mid \mathcal{A})$ and $\log P(s)$ depends only on s , the restriction of $\log V$ to Π_+^A is strictly supermodular in (s, θ) .

Assume φ is an equilibrium. By Topkis's theorem, φ is weakly increasing. Now we show that it is strictly increasing. Fix $\theta_1 < \theta_2$ and suppose $\varphi(\theta) = s_0$ for each $\theta \in [\theta_1, \theta_2]$. Since $\mathcal{A}$ is well-behaved and $U(s_0, \theta \mid \mathcal{A}) \geq 0$ for each type θ , it follows that $U(s_0, \theta_1 \mid \mathcal{A}) > 0$. Moreover, any score $s < s_0$ discontinuously raises the probability of winning. As U is continuous in the score, it is strictly profitable for θ_1 to deviate to some s sufficiently close to s_0 , leading to a contradiction. Hence, φ is strictly increasing.

Next, we show that φ is continuous. If it jumped at θ_0 , with limits $s_- < s_+$, then each score $s \in (s_-, s_+)$ would beat exactly the same opponents as s_- , so $P(s) = P(s_-)$. Since U is strictly increasing in s , s_- would then be strictly dominated. Thus, φ is continuous.

We now show that φ is differentiable. Define $\psi(\theta', \theta) := (1 - F(\theta'))^{I-1} U(\varphi(\theta'), \theta \mid \mathcal{A})$, so that $\theta \in \arg \max_{\theta'} \psi(\theta', \theta)$. For $\theta' > \theta$, optimality implies

$$(1 - F(\theta'))^{I-1} U(\varphi(\theta'), \theta \mid \mathcal{A}) \leq (1 - F(\theta))^{I-1} U(\varphi(\theta), \theta \mid \mathcal{A}).$$

By the mean value theorem, there exists $\tilde{s}$ between $\varphi(\theta)$ and $\varphi(\theta')$ such that

$$U(\varphi(\theta'), \theta \mid \mathcal{A}) - U(\varphi(\theta), \theta \mid \mathcal{A}) = U_1(\tilde{s}, \theta \mid \mathcal{A})(\varphi(\theta') - \varphi(\theta)).$$

Substituting and rearranging yields

$$\frac{\varphi(\theta') - \varphi(\theta)}{\theta' - \theta} \leq \frac{U(\varphi(\theta), \theta \mid \mathcal{A})[(1 - F(\theta))^{I-1} - (1 - F(\theta'))^{I-1}]}{(\theta' - \theta)(1 - F(\theta'))^{I-1} U_1(\tilde{s}, \theta \mid \mathcal{A})}.$$

Reversing the roles of θ and θ' yields a corresponding lower bound. Since $F' = f$, continuity of φ , f , U , and U_1 , together with $U_1 > 0$, implies that both bounds converge to the same limit as $\theta' \rightarrow \theta$ at every interior type θ . Hence φ is differentiable on $(\underline{\theta}, \bar{\theta})$. The differential equation derived below has a continuous right-hand side on the interior, so φ is continuously differentiable there.

We now characterize φ . Since φ is strictly increasing, a type- θ seller who submits a bid s wins with probability $(1 - F(\varphi^{-1}(s)))^{I-1}$. Her expected utility is therefore $V(s, \theta) = (1 - F(\varphi^{-1}(s)))^{I-1} U(s, \theta \mid \mathcal{A})$. In equilibrium, $s = \varphi(\theta)$ is optimal, so $\partial V(s, \theta) / \partial s|_{s=\varphi(\theta)} = 0$. Differentiating,

$$\begin{aligned} \frac{\partial V(s, \theta)}{\partial s} &= -(I-1) (1 - F(\varphi^{-1}(s)))^{I-2} f(\varphi^{-1}(s)) \frac{1}{\varphi'(\varphi^{-1}(s))} U(s, \theta \mid \mathcal{A}) \\ &\quad + (1 - F(\varphi^{-1}(s)))^{I-1} U_1(s, \theta \mid \mathcal{A}). \end{aligned}$$

Evaluating at $s = \varphi(\theta)$ and rearranging gives

$$\varphi'(\theta) = \frac{(I-1)f(\theta)}{1 - F(\theta)} \frac{U(\varphi(\theta), \theta \mid \mathcal{A})}{U_1(\varphi(\theta), \theta \mid \mathcal{A})}. \quad (6)$$

The boundary condition $U(\varphi(\bar{\theta}), \bar{\theta} \mid \mathcal{A}) = 0$ follows from the participation constraint of the highest type. For every $\theta < \bar{\theta}$, equilibrium requires $U(\varphi(\theta), \theta \mid \mathcal{A}) > 0$, so continuity implies $U(\varphi(\bar{\theta}), \bar{\theta} \mid \mathcal{A}) \geq 0$. Conversely, if $U(\varphi(\bar{\theta}), \bar{\theta} \mid \mathcal{A}) > 0$, then, by continuity, the highest type could profitably lower her bid slightly, obtaining a positive probability of winning while retaining positive utility conditional on winning. Hence, the boundary condition must bind.

To establish existence, we show that Equation (6) admits a solution satisfying the boundary condition $U(\varphi(\bar{\theta}), \bar{\theta} \mid \mathcal{A}) = 0$. Since the auction is rich, for each θ there exists a unique

score $r(\theta) \in \mathcal{S}$ such that $U(r(\theta), \theta \mid \mathcal{A}) = 0$. Continuity and well-behavedness imply that r is continuous and strictly increasing. For each $\tau \in (\underline{\theta}, \bar{\theta})$, let φ_τ denote a solution to the equilibrium differential equation obtained by integrating backward from the terminal condition $\varphi_\tau(\tau) = r(\tau)$. Since $1 - F(\tau) > 0$, Peano's theorem guarantees the existence of such a solution locally below τ .

Moreover, $r(\theta) \leq \varphi_\tau(\theta) \leq r(\tau)$ for every $\theta \leq \tau$. To see this, note that $U(r(\theta), \theta \mid \mathcal{A}) = 0$ and, by well-behavedness, r is strictly increasing. At any point where $\varphi_\tau(\theta) = r(\theta)$, the right-hand side of the differential equation vanishes. The scalar comparison principle therefore prevents a solution integrated backward from τ from crossing below r . Since the differential equation is nonnegative whenever $\varphi_\tau(\theta) \geq r(\theta)$, φ_τ is nondecreasing, yielding the upper bound $\varphi_\tau(\theta) \leq r(\tau)$. These bounds keep the solution in $\mathcal{S}$ and, by continuity of the right-hand side of the differential equation on compact intervals bounded away from $\bar{\theta}$, allow it to extend to $\underline{\theta}$.

It remains to extend the argument to the limit as τ converges to $\bar{\theta}$. On every compact subinterval of $[\underline{\theta}, \bar{\theta})$, the family $\{\varphi_\tau\}$ is uniformly bounded and, by (6), equicontinuous. Hence, by the Arzelà–Ascoli theorem, there exists a subsequence converging locally uniformly to a solution φ of (6) on $[\underline{\theta}, \bar{\theta})$. The bounds above pass to the limit, implying that $r(\theta) \leq \varphi(\theta) \leq r(\bar{\theta})$. Hence, since r is continuous, $\lim_{\theta \uparrow \bar{\theta}} \varphi(\theta) = r(\bar{\theta})$. Defining $\varphi(\bar{\theta}) = r(\bar{\theta})$ therefore extends the solution to $[\underline{\theta}, \bar{\theta}]$ and satisfies $U(\varphi(\bar{\theta}), \bar{\theta} \mid \mathcal{A}) = 0$, as required.

We show $\varphi(\theta) > r(\theta)$ for each type $\theta < \bar{\theta}$. Since $\varphi \geq r$, it suffices to show that no interior type θ_0 satisfies $\varphi(\theta_0) = r(\theta_0)$. At any such type, well-behavedness gives $U(r(\theta_0), \theta \mid \mathcal{A}) < 0$ for each $\theta > \theta_0$, and concavity in the score and the strict positivity of U_1 imply, through the equilibrium differential equation, that $0 \leq \varphi'(\theta) \leq C[\varphi(\theta) - r(\theta_0)]$ to the right of θ_0 , for

some finite constant C . Since φ is nondecreasing, for each $\theta \in (\theta_0, \theta_0 + 1/C)$,

$$0 \leq \varphi(\theta) - r(\theta_0) = \int_{\theta_0}^{\theta} \varphi'(t) dt \leq C \int_{\theta_0}^{\theta} [\varphi(t) - r(\theta_0)] dt \leq C(\theta - \theta_0)[\varphi(\theta) - r(\theta_0)],$$

so $\varphi(\theta) = r(\theta_0) < r(\theta)$, which is incompatible with $\varphi \geq r$. Hence $\varphi(\theta) > r(\theta)$, so $U(\varphi(\theta), \theta \mid \mathcal{A}) > 0$ and $\varphi'(\theta) > 0$ at each interior type.

It remains to verify global optimality. Parameterize a deviation by the type θ' whose equilibrium bid is mimicked and define $\psi(\theta', \theta) := (1 - F(\theta'))^{I-1} U(\varphi(\theta'), \theta \mid \mathcal{A})$. Since equilibrium winning utility is strictly positive, deviations yielding nonpositive utility cannot be profitable. We therefore restrict attention to deviations for which $\psi(\theta', \theta) > 0$. For such deviations with $\theta' \in (\underline{\theta}, \bar{\theta})$,

$$\frac{\partial \log \psi(\theta', \theta)}{\partial \theta'} = -\frac{(I-1)f(\theta')}{1-F(\theta')} + \frac{U_1(\varphi(\theta'), \theta \mid \mathcal{A})}{U(\varphi(\theta'), \theta \mid \mathcal{A})} \varphi'(\theta').$$

Using the equilibrium differential equation evaluated at θ' gives

$$\frac{\partial \log \psi(\theta', \theta)}{\partial \theta'} = \frac{(I-1)f(\theta')}{1-F(\theta')} \left[\frac{\frac{U_1(\varphi(\theta'), \theta \mid \mathcal{A})}{U(\varphi(\theta'), \theta \mid \mathcal{A})}}{\frac{U_1(\varphi(\theta'), \theta' \mid \mathcal{A})}{U(\varphi(\theta'), \theta' \mid \mathcal{A})}} - 1 \right].$$

Since U is strictly log-supermodular on $\Pi_+^{\mathcal{A}}$, $\frac{U_1(s, \theta \mid \mathcal{A})}{U(s, \theta \mid \mathcal{A})} = \frac{\partial \log U(s, \theta \mid \mathcal{A})}{\partial s}$ is strictly increasing in θ .

Therefore,

$$\frac{\partial \log \psi(\theta', \theta)}{\partial \theta'} \begin{cases} > 0, & \theta' < \theta, \\ = 0, & \theta' = \theta, \\ < 0, & \theta' > \theta. \end{cases}$$

Consequently, φ is a symmetric equilibrium. We now establish uniqueness of its continuous

bidding strategy.

Suppress the auction argument and write $R(s, \theta) := U(s, \theta)/U_1(s, \theta)$. For $s_2 > s_1$ with $U(s_1, \theta) > 0$, concavity implies $U_1(s_2, \theta) \leq U_1(s_1, \theta)$ and

$$U(s_2, \theta) - U(s_1, \theta) \geq U_1(s_2, \theta)(s_2 - s_1).$$

Hence

$$\begin{aligned} R(s_2, \theta) - R(s_1, \theta) &= \frac{U(s_2, \theta) - U(s_1, \theta)}{U_1(s_2, \theta)} \\ &\quad + U(s_1, \theta) \left[\frac{1}{U_1(s_2, \theta)} - \frac{1}{U_1(s_1, \theta)} \right] \geq s_2 - s_1. \end{aligned}$$

Let φ_1 and φ_2 be two continuous feasible solutions of the equilibrium differential equation with $U(\varphi_j(\theta), \theta) > 0$ at every interior type, $j = 1, 2$. Suppose $\delta(\theta_0) := \varphi_2(\theta_0) - \varphi_1(\theta_0) > 0$ at some interior type θ_0 . Wherever $\delta > 0$,

$$\begin{aligned} \delta'(\theta) &= \frac{(I-1)f(\theta)}{1-F(\theta)} [R(\varphi_2(\theta), \theta) - R(\varphi_1(\theta), \theta)] \\ &\geq \frac{(I-1)f(\theta)}{1-F(\theta)} \delta(\theta). \end{aligned}$$

The gap therefore remains positive to the right of θ_0 . Integrating gives

$$\delta(\theta) \geq \delta(\theta_0) \left[\frac{1-F(\theta_0)}{1-F(\theta)} \right]^{I-1} \quad \text{for } \theta_0 \leq \theta < \bar{\theta}.$$

The right-hand side diverges as θ converges to $\bar{\theta}$, contradicting $\varphi_1(\theta), \varphi_2(\theta) \in S$ and compactness of S . Interchanging the roles of φ_1 and φ_2 establishes equality throughout the interior. Continuity extends this equality to the endpoints. Thus the continuous equilibrium bidding

strategy is unique.

Log-submodular case. Consider the transformation $\omega = \bar{\theta} + \underline{\theta} - \theta$, which reverses the ordering of types while preserving the support $[\underline{\theta}, \bar{\theta}]$. Its distribution and density are $G(\omega) = 1 - F(\bar{\theta} + \underline{\theta} - \omega)$ and $g(\omega) = f(\bar{\theta} + \underline{\theta} - \omega)$. Define $\tilde{U}(s, \omega \mid \mathcal{A}) = U(s, \bar{\theta} + \underline{\theta} - \omega \mid \mathcal{A})$. Because $\log U$ has strictly decreasing differences in (s, θ) , $\log \tilde{U}$ has strictly increasing differences in (s, ω) . Thus, the transformed environment is well behaved, and the preceding argument applies.

The unique symmetric equilibrium bidding strategy $\tilde{\varphi}(\omega)$ therefore satisfies

$$\tilde{\varphi}'(\omega) = \frac{(I-1)g(\omega)}{1-G(\omega)} \frac{\tilde{U}(\tilde{\varphi}(\omega), \omega \mid \mathcal{A})}{\tilde{U}_1(\tilde{\varphi}(\omega), \omega \mid \mathcal{A})},$$

with boundary condition $\tilde{U}(\tilde{\varphi}(\bar{\theta}), \bar{\theta} \mid \mathcal{A}) = 0$. Since $g(\omega) = f(\theta)$, $1 - G(\omega) = F(\theta)$, $\tilde{U}(s, \omega \mid \mathcal{A}) = U(s, \theta \mid \mathcal{A})$, and $\varphi'(\theta) = -\tilde{\varphi}'(\omega)$, the equilibrium bidding strategy in the original environment satisfies

$$\varphi'(\theta) = -\frac{(I-1)f(\theta)}{F(\theta)} \frac{U(\varphi(\theta), \theta \mid \mathcal{A})}{U_1(\varphi(\theta), \theta \mid \mathcal{A})},$$

with boundary condition $U(\varphi(\underline{\theta}), \underline{\theta} \mid \mathcal{A}) = 0$, since the highest type in the transformed environment corresponds to the lowest type in the original environment.

Proof of Theorem 1

Before proving Theorem 1, we establish three auxiliary lemmas that will be useful in characterizing whether an auction is well behaved or induces perverse incentives. The first provides a sufficient condition for strict log-supermodularity, while the remaining two characterize the relevant monotonicity conditions used in the analysis of scaling auctions.

Lemma 1. *If $\frac{\partial^2 U}{\partial s \partial \theta} \geq 0$, $\frac{\partial U}{\partial s} > 0$, and $\frac{\partial U}{\partial \theta} < 0$, then $U : \Pi_+^A \rightarrow \mathbb{R}$ is strictly log-supermodular.*

Proof. Direct calculation gives

$$\frac{\partial^2 \log U}{\partial s \partial \theta} = \frac{\frac{\partial^2 U}{\partial s \partial \theta} U - \frac{\partial U}{\partial s} \frac{\partial U}{\partial \theta}}{U^2}.$$

Since $U > 0$ on Π_+^A , the denominator is positive. Moreover, if $\frac{\partial^2 U}{\partial s \partial \theta} \geq 0$, $\frac{\partial U}{\partial s} > 0$, and $\frac{\partial U}{\partial \theta} < 0$, the numerator is strictly positive. Hence, U is strictly log-supermodular on Π_+^A . $\square$

Lemma 2. *Let $a : [\underline{\theta}, \bar{\theta}] \rightarrow \mathbb{R}_{++}$ and $c : [\underline{\theta}, \bar{\theta}] \rightarrow \mathbb{R}_{++}$ be continuous mappings. If the ratio $a(\theta)/c(\theta)$ is strictly increasing (resp. decreasing), then $U(s, \theta) := sa(\theta) - c(\theta)$ is strictly log-submodular (resp. log-supermodular) on $\{(s, \theta) : U(s, \theta) > 0\}$.*

Proof. Fix $s' > s$ and $\theta' > \theta$ such that $U(s', \theta)$, $U(s, \theta')$, $U(s', \theta')$, and $U(s, \theta)$ are all strictly positive. Expanding the products gives

$$\begin{aligned} U(s', \theta)U(s, \theta') &= (s'a(\theta) - c(\theta))(sa(\theta') - c(\theta')) \\ &= s'sa(\theta)a(\theta') - s'a(\theta)c(\theta') - sc(\theta)a(\theta') + c(\theta)c(\theta'). \end{aligned}$$

Likewise, $U(s', \theta')U(s, \theta) = s'sa(\theta')a(\theta) - s'a(\theta')c(\theta) - sc(\theta')a(\theta) + c(\theta')c(\theta)$. Subtracting and canceling common terms yields

$$\begin{aligned} U(s', \theta)U(s, \theta') - U(s', \theta')U(s, \theta) &= (s' - s)(a(\theta')c(\theta) - a(\theta)c(\theta')) \\ &= (s' - s)c(\theta)c(\theta') \left(\frac{a(\theta')}{c(\theta')} - \frac{a(\theta)}{c(\theta)} \right). \end{aligned}$$

Since $(s' - s)c(\theta)c(\theta') > 0$, the sign of the difference is determined by the last term. So,

if $a(\theta)/c(\theta)$ is strictly increasing (resp. decreasing), U is strictly log-submodular (resp. log-supermodular). $\square$

Lemma 3. Write $\overline{Q}_t(\theta) = q_t(\theta) + \mathbb{E}[\xi_t^q]$. A task $t \in \mathcal{T}$ is elastic (resp. inelastic) if and only if the ratio $\overline{Q}_t(\theta)/\text{EC}(\theta)$ is strictly increasing (resp. decreasing) in θ .

Proof. Observe that

$$\begin{aligned} \frac{d}{d\theta} \left(\frac{\overline{Q}_t(\theta)}{\text{EC}(\theta)} \right) &= \frac{\overline{Q}_t'(\theta)\text{EC}(\theta) - \overline{Q}_t(\theta)\text{EC}'(\theta)}{\text{EC}(\theta)^2} \\ &= \frac{\overline{Q}_t(\theta)}{\text{EC}(\theta)} \left(\frac{\overline{Q}_t'(\theta)}{\overline{Q}_t(\theta)} - \frac{\text{EC}'(\theta)}{\text{EC}(\theta)} \right). \end{aligned}$$

Since $\overline{Q}_t(\theta)/\text{EC}(\theta) > 0$, the derivative has the same sign as the last term. So, t is elastic (resp. inelastic) if and only if $\overline{Q}_t(\theta)/\text{EC}(\theta)$ is strictly increasing (resp. decreasing) in θ . $\square$

Having established these lemmata we now prove Theorem 1. We proceed by verifying, for each auction format, the conditions under which the auction is well behaved or induces perverse incentives. The profit of a type- θ seller with score s , shock $\boldsymbol{\xi}$, and alternative b is

$$\pi(s, \theta, b, \boldsymbol{\xi}) := \Pi(\mathbf{O}(\theta, \boldsymbol{\xi}), s, b) - (\mathbf{q}(\theta) + \boldsymbol{\xi}^q) \cdot (\mathbf{c}(\theta) + \boldsymbol{\xi}^c) - k(\theta) - \xi^k.$$

The corresponding interim utility is $U(s, \theta \mid \mathcal{A}) := \max_{b \in \mathcal{B}(s)} \mathbb{E} [u(\pi(s, \theta, b, \boldsymbol{\xi}))]$. Throughout, we analyze the properties of $U(s, \theta \mid \mathcal{A})$ on the domain of positive utility, $\Pi_+^{\mathcal{A}}$.

Fixed-payment auction. In a fixed-payment auction, $\Pi(s) = s$, leading to net profit $\pi = s - (\mathbf{q}(\theta) + \boldsymbol{\xi}^q) \cdot (\mathbf{c}(\theta) + \boldsymbol{\xi}^c) - k(\theta) - \xi^k$. Observe that $\frac{\partial \pi}{\partial s} = 1$ and, by assumption, $\frac{\partial \pi}{\partial \theta} < 0$. Moreover, the cross-derivative of profit is $\frac{\partial^2 \pi}{\partial s \partial \theta} = 0$. The interim utility is $U(s, \theta \mid \mathcal{A}) = \mathbb{E}[u(\pi)]$, where $\frac{\partial U}{\partial s} = \mathbb{E}[u'(\pi)] > 0$ and $\frac{\partial U}{\partial \theta} = \mathbb{E}[u'(\pi) \frac{\partial \pi}{\partial \theta}] < 0$. As the cross derivative of profit $\frac{\partial^2 \pi}{\partial s \partial \theta}$ is zero,

the cross-derivative $\frac{\partial^2 U}{\partial s \partial \theta} = \mathbb{E}[u''(\pi) \frac{\partial \pi}{\partial \theta}] \geq 0$, since $u'' \leq 0$ and $\frac{\partial \pi}{\partial \theta} < 0$. So, by Lemma 1, U is strictly log-supermodular on Π_+^A .

Cost-sharing auction. The realized marginal cost $\mathbf{O}(\theta, \boldsymbol{\xi}) := (\mathbf{q}(\theta) + \boldsymbol{\xi}^q) \cdot (\mathbf{c}(\theta) + \boldsymbol{\xi}^c)$ is weakly increasing in θ by assumption. The payment rule is $\Pi(\mathbf{O}(\theta, \boldsymbol{\xi}), s) = X + s \mathbf{O}(\theta, \boldsymbol{\xi})$. Profit is $\pi = X - k(\theta) - \xi^k + (s - 1) \mathbf{O}(\theta, \boldsymbol{\xi})$. The derivatives are $\frac{\partial \pi}{\partial s} = \mathbf{O}(\theta, \boldsymbol{\xi}) > 0$ and $\frac{\partial \pi}{\partial \theta} = -k'(\theta) - (1 - s) \frac{\partial \mathbf{O}}{\partial \theta} < 0$. The cross-derivative is $\frac{\partial^2 \pi}{\partial s \partial \theta} = \frac{\partial \mathbf{O}}{\partial \theta} \geq 0$. For interim utility $U(s, \theta \mid \mathcal{A}) = \mathbb{E}[u(\pi)]$, we find $\frac{\partial U}{\partial s} = \mathbb{E}[u'(\pi) \frac{\partial \pi}{\partial s}] > 0$ and $\frac{\partial U}{\partial \theta} = \mathbb{E}[u'(\pi) \frac{\partial \pi}{\partial \theta}] < 0$. The cross-derivative satisfies $\frac{\partial^2 U}{\partial s \partial \theta} = \mathbb{E}[u''(\pi) \frac{\partial \pi}{\partial \theta} \frac{\partial \pi}{\partial s} + u'(\pi) \frac{\partial^2 \pi}{\partial s \partial \theta}] \geq 0$. Lemma 1 shows U is strictly log-supermodular on Π_+^A .

Scaling auction with risk neutrality and homogeneous quantities. As quantities are homogeneous, there is some $\mathbf{q}^*$ such that $\mathbf{q}(\theta) = \mathbf{q}^*$ for each θ . Since the payment rule is $\Pi(\mathbf{O}(\theta, \boldsymbol{\xi}), s, \mathbf{b}) = (\mathbf{q}^* + \boldsymbol{\xi}^q) \cdot \mathbf{b}$, and the agent is risk neutral, each type selects a revenue maximizing vector that does not depend on θ . Write $\mathbf{b}^*(s)$ for the vector that maximizes expected revenue under score s . By risk neutrality, $\mathbf{b}^*(s)$ is a corner solution that is linear in s . The profit of type θ with score s is $\pi = (\mathbf{q}^* + \boldsymbol{\xi}^q) \cdot \mathbf{b}^*(s) - (\mathbf{q}^q + \boldsymbol{\xi}^q) \cdot (\mathbf{c}(\theta) + \boldsymbol{\xi}^c) - k(\theta)$. Since the corner solution $\mathbf{b}^*(s)$ is linear in s and the agent is risk neutral, the expected utility is of the form $U(s, \theta \mid \mathcal{A}) = sR - EC(\theta)$, for some constant $R > 0$ and expected cost $EC(\theta)$. Since U is affine in s , the cross-derivative of $\log U$ satisfies

$$\frac{\partial^2 \log U}{\partial s \partial \theta} = \frac{\partial}{\partial \theta} \left(\frac{R}{sR - EC(\theta)} \right) = \frac{R EC'(\theta)}{(sR - EC(\theta))^2} > 0,$$

on the domain of positive utility. This implies that U is strictly log-supermodular on Π_+^A .

Scaling Auction under risk neutrality. Write $\overline{\mathbf{Q}}(\theta) := \mathbb{E}[\mathbf{q}(\theta) + \boldsymbol{\xi}^q \mid \theta]$, for the expected

quantities of type θ . Under risk neutrality, conditional on winning at a score s , type θ solves

$$\max_{\mathbf{b} \geq 0} \overline{\mathbf{Q}}(\theta) \cdot \mathbf{b} - \text{EC}(\theta) \quad \text{s.t.} \quad \mathbf{w} \cdot \mathbf{b} = s.$$

where, $\text{EC}(\theta) := \overline{\mathbf{Q}}(\theta) \cdot \mathbf{c}(\theta) + k(\theta)$. As the objective is linear in $\mathbf{b}$, the optimum is attained at a corner. Hence, there exists some task t maximizing $\overline{Q}_t(\theta)/w_t$ such that $b_t = s/w_t$ and $b_\tau = 0$ for all $\tau \neq t$. Consequently,

$$U(s, \theta \mid \mathcal{A}_S) = \max_{t \in \mathcal{T}} s \frac{\overline{Q}_t(\theta)}{w_t} - \text{EC}(\theta) = sR(\theta) - \text{EC}(\theta),$$

where $R(\theta) = \max_{t \in \mathcal{T}} \overline{Q}_t(\theta)/w_t$. By Lemma 3, the fact that each task t is elastic (resp. inelastic) implies that $\overline{Q}_t(\theta)/\text{EC}(\theta)$ is strictly increasing (resp. decreasing) in θ . Moreover, since $w_t > 0$, $\rho_t(\theta) := \frac{1}{w_t} \frac{\overline{Q}_t(\theta)}{\text{EC}(\theta)}$ inherits the same strict monotonicity for each $t \in \mathcal{T}$. Since $R(\theta)/\text{EC}(\theta) = \max_{t \in \mathcal{T}} \rho_t(\theta)$ and the maximum of finitely many strictly increasing (resp. decreasing) functions is itself strictly increasing (resp. decreasing), it follows that $R(\theta)/\text{EC}(\theta)$ is strictly increasing (resp. decreasing) in θ . Apply Lemma 2 with $a(\theta) = R(\theta)$ and $c(\theta) = \text{EC}(\theta)$ to conclude that $U(s, \theta \mid \mathcal{A}_S) = sR(\theta) - \text{EC}(\theta)$ is strictly log-submodular (resp. log-supermodular) on $\Pi_+^{\mathcal{A}_S}$.

Proof of Theorem 2

Before proving Theorem 2, we establish two auxiliary lemmas. The first compares procurement costs under efficient and perverse allocations. The second ranks two well-behaved mechanisms according to the informational rents implied by their marginal utilities with respect to type.

Lemma 4. *Consider a risk-neutral environment and let $\theta^{(n)}$ denote the n -th order statistic of the bidder types, where $\theta^{(1)}$ is the lowest type and $\theta^{(I)}$ the highest type. The expected procurement cost in the fixed-payment auction satisfies $\mathbb{E}[C^{\mathcal{A}_{FP}}] = \mathbb{E}[\text{EC}(\theta^{(2)})]$. Moreover, if an auction $\mathcal{A}$ induces perverse incentives, then*

$$\mathbb{E}[C^{\mathcal{A}}] \geq \mathbb{E}[\text{EC}(\theta^{(I)})] \geq \mathbb{E}[\text{EC}(\theta^{(2)})] = \mathbb{E}[C^{\mathcal{A}_{FP}}].$$

Proof. By Proposition 1, an auction with perverse incentives has a decreasing equilibrium score strategy. Since the lowest score wins, the project is awarded to the highest type, $\theta^{(I)}$. Under risk neutrality, the buyer's expected payment to the winning seller equals expected production cost plus informational rent. Since equilibrium rents are nonnegative, $\mathbb{E}[C^{\mathcal{A}}] \geq \mathbb{E}[\text{EC}(\theta^{(I)})]$.

We next derive the benchmark $\mathcal{A}_{FP}$. Under risk neutrality, $U(s, \theta \mid \mathcal{A}_{FP}) = s - \text{EC}(\theta)$ and $U_1(s, \theta \mid \mathcal{A}_{FP}) = 1$. Since the fixed-payment auction is well behaved, Proposition 1 implies $\varphi'_{\mathcal{A}_{FP}}(\theta) = \frac{(I-1)f(\theta)}{1-F(\theta)} [\varphi_{\mathcal{A}_{FP}}(\theta) - \text{EC}(\theta)]$. Equivalently,

$$\frac{d}{d\theta} ([1 - F(\theta)]^{I-1} \varphi_{\mathcal{A}_{FP}}(\theta)) = -(I-1)f(\theta)[1 - F(\theta)]^{I-2} \text{EC}(\theta).$$

Integrating from θ to $\bar{\theta}$ gives

$$[1 - F(\theta)]^{I-1} \varphi_{\mathcal{A}_{FP}}(\theta) = (I-1) \int_{\theta}^{\bar{\theta}} f(t)[1 - F(t)]^{I-2} \text{EC}(t) dt.$$

The fixed-payment auction allocates the project to $\theta^{(1)}$, whose density is $If(\theta)[1 -$

$F(\theta)]^{I-1}$. Therefore,

$$\begin{aligned}\mathbb{E}[C^{\mathcal{A}_{FP}}] &= I(I-1) \int_{\underline{\theta}}^{\bar{\theta}} f(\theta) \int_{\theta}^{\bar{\theta}} f(t)[1-F(t)]^{I-2} \text{EC}(t) dt d\theta \\ &= I(I-1) \int_{\underline{\theta}}^{\bar{\theta}} F(t)[1-F(t)]^{I-2} f(t) \text{EC}(t) dt \\ &= \mathbb{E}[\text{EC}(\theta^{(2)})],\end{aligned}$$

where the second equality changes the order of integration and the last uses the density of the second-order statistic. Finally, since $\theta^{(2)} \leq \theta^{(I)}$ almost surely and EC is increasing, $\mathbb{E}[\text{EC}(\theta^{(2)})] \leq \mathbb{E}[\text{EC}(\theta^{(I)})]$. Combining these inequalities proves the result. $\square$

Lemma 5. *Consider two well-behaved mechanisms $\mathcal{A}$ and $\mathcal{A}'$ in the same risk-neutral environment. If $U_{\theta}(\varphi_{\mathcal{A}}(\theta), \theta \mid \mathcal{A}) \geq U_{\theta}(\varphi_{\mathcal{A}'}(\theta), \theta \mid \mathcal{A}')$ almost surely among the set of types, then $\mathbb{E}[C^{\mathcal{A}}] \leq \mathbb{E}[C^{\mathcal{A}'}]$. If the inequality is strict for all $\theta \in (\underline{\theta}, \bar{\theta})$, then $\mathbb{E}[C^{\mathcal{A}}] < \mathbb{E}[C^{\mathcal{A}'}]$.*

Proof. Since both mechanisms are well behaved, both allocate the project efficiently. Hence, a type θ wins with probability $P(\theta) := [1 - F(\theta)]^{I-1}$. Let $V_{\mathcal{A}}(\theta) := P(\theta) U(\varphi_{\mathcal{A}}(\theta), \theta \mid \mathcal{A})$ denote the seller's interim expected profit before opponents' types are realized. For any score s in the range of $\varphi_{\mathcal{A}}$, define $\hat{V}_{\mathcal{A}}(s, \theta) := [1 - F(\varphi_{\mathcal{A}}^{-1}(s))]^{I-1} U(s, \theta \mid \mathcal{A})$. So, in equilibrium, $V_{\mathcal{A}}(\theta) = \max_s \hat{V}_{\mathcal{A}}(s, \theta)$. By the envelope theorem, $V'_{\mathcal{A}}(\theta) = P(\theta) U_{\theta}(\varphi_{\mathcal{A}}(\theta), \theta \mid \mathcal{A})$, and, using the boundary condition $V_{\mathcal{A}}(\bar{\theta}) = 0$,

$$V_{\mathcal{A}}(\theta) = \int_{\theta}^{\bar{\theta}} P(t) [-U_{\theta}(\varphi_{\mathcal{A}}(t), t \mid \mathcal{A})] dt.$$

Thus, $U_{\theta}(\varphi_{\mathcal{A}}(\theta), \theta \mid \mathcal{A}) \geq U_{\theta}(\varphi_{\mathcal{A}'}(\theta), \theta \mid \mathcal{A}')$ implies $V_{\mathcal{A}}(\theta) \leq V_{\mathcal{A}'}(\theta)$ for every type, with strict inequality whenever the comparison of marginal utilities is strict on the interior.

Because the two mechanisms induce the same efficient allocation, their expected production costs coincide, and expected procurement costs differ only through informational rents:

$$\mathbb{E}[C^{\mathcal{A}}] = \mathbb{E}[\text{EC}(\theta^{(1)})] + I \int_{\underline{\theta}}^{\bar{\theta}} V_{\mathcal{A}}(\theta) dF(\theta).$$

The stated cost ranking follows. $\square$

Having established these two auxiliary results, we now prove Theorem 2. For an equilibrium strategy $\varphi_{\mathcal{A}}$, let $C^{\mathcal{A}}(\theta)$ denote the buyer's expected payment conditional on a type- θ seller winning. Fixed-payment and cost-sharing auctions are well behaved, so the winner has type $\theta^{(1)}$. A scaling auction is evaluated at $\theta^{(1)}$ when it is well behaved and at $\theta^{(I)}$ when it induces perverse incentives.

Accordingly,

$$\begin{aligned} \mathbb{E}[C^{\mathcal{A}_{FP}}] &= \mathbb{E}[\varphi_{\mathcal{A}_{FP}}(\theta^{(1)})], \\ \mathbb{E}[C^{\mathcal{A}_{CS}}] &= X + \mathbb{E}[\varphi_{\mathcal{A}_{CS}}(\theta^{(1)}) (\mathbf{c}(\theta^{(1)}) + \boldsymbol{\xi}^c) \cdot (\mathbf{q}(\theta^{(1)}) + \boldsymbol{\xi}^q)], \\ \mathbb{E}[C^{\mathcal{A}_S}] &= \mathbb{E}[(\mathbf{q}(\theta^{(\iota)}) + \boldsymbol{\xi}^q) \cdot \mathbf{b}_{\mathcal{A}_S}(\theta^{(\iota)})], \end{aligned}$$

where $\iota = 1$ when the scaling auction is well behaved and $\iota = I$ when it induces perverse incentives. The two lemmas isolate the two forces relevant for the comparison: Lemma 4 captures the allocation effect, whereas Lemma 5 compares informational rents when mechanisms induce the same efficient allocation.

Types as fixed costs. Because type enters only through fixed costs, all mechanisms induce the same certainty equivalent per score net of $k(\theta)$ for each type. Under risk neutrality, certainty equivalents coincide with expected payments, so all mechanisms leave each type the

same informational rent. Since they also induce the same allocation, expected procurement costs are identical across fixed-payment, scaling, and cost-sharing auctions.

Types as marginal costs. Input quantities and fixed costs are homogeneous, while marginal costs depend on type. Write $K = k + \mathbb{E}[\xi^k]$ for expected fixed costs and $M(\theta) := \mathbb{E}[(\mathbf{c}(\theta) + \boldsymbol{\xi}^c) \cdot (\mathbf{q} + \boldsymbol{\xi}^q)]$ for expected marginal costs. So $\text{EC}(\theta) = k + M(\theta)$. Under fixed payment, $U(s, \theta \mid \mathcal{A}_{FP}) = s - K - M(\theta)$ and $U_\theta(s, \theta \mid \mathcal{A}_{FP}) = -M'(\theta)$. Under cost sharing, $U(s, \theta \mid \mathcal{A}_{CS}) = X - K - (1-s)M(\theta)$ and $U_\theta(s, \theta \mid \mathcal{A}_{CS}) = -(1-s)M'(\theta)$. Hence, for every $s \in [0, 1]$, $U_\theta(s, \theta \mid \mathcal{A}_{CS}) \geq U_\theta(s', \theta \mid \mathcal{A}_{FP})$ for any fixed-payment score s' , with strict inequality whenever $s > 0$ and $M'(\theta) > 0$. Lemma 5 therefore implies $\mathbb{E}[C^{\mathcal{A}_{CS}}] < \mathbb{E}[C^{\mathcal{A}_{FP}}]$.

It remains to compare scaling and fixed payment. Since $\mathbf{q}$ is homogeneous, a scaling score s induces expected utility $U(s, \theta \mid \mathcal{A}_S) = sR - K - M(\theta)$ for some constant $R > 0$ determined by expected quantities and the scoring weights. After the normalization $\tilde{s} = sR$, $U(\tilde{s}, \theta \mid \mathcal{A}_S) = \tilde{s} - K - M(\theta) = U(\tilde{s}, \theta \mid \mathcal{A}_{FP})$. Thus, scaling and fixed payment generate the same informational rents. Therefore, $\mathbb{E}[C^{\mathcal{A}_S}] = \mathbb{E}[C^{\mathcal{A}_{FP}}]$.

Types as quantities. In the types-as-quantities environment, $k(\theta) = k$ and $\mathbf{c}(\theta) = \mathbf{c}$ are constant. Write $K := k + \mathbb{E}[\xi^k]$ and $M(\theta) := \mathbb{E}[(\mathbf{q}(\theta) + \boldsymbol{\xi}^q) \cdot (\mathbf{c} + \boldsymbol{\xi}^c)]$. Then $\text{EC}(\theta) := K + M(\theta)$ is strictly increasing in θ . By Lemma 4, $\mathbb{E}[C^{\mathcal{A}_{FP}}] = \mathbb{E}[\text{EC}(\theta^{(2)})]$.

We first compare cost sharing and fixed payment. Under fixed payment, $U(s, \theta \mid \mathcal{A}_{FP}) = s - k - M(\theta)$ and $U_\theta(s, \theta \mid \mathcal{A}_{FP}) = -M'(\theta)$, whereas under cost sharing, $U(s, \theta \mid \mathcal{A}_{CS}) = X - k - (1-s)M(\theta)$ and $U_\theta(s, \theta \mid \mathcal{A}_{CS}) = -(1-s)M'(\theta)$. Thus, for every cost-sharing score $s \in [0, 1]$, $U_\theta(s, \theta \mid \mathcal{A}_{CS}) \geq U_\theta(s, \theta \mid \mathcal{A}_{FP})$, with strict inequality whenever $s > 0$. Since all types above $\underline{\theta}$ bid strictly above zero, Lemma 5 implies that $\mathbb{E}[C^{\mathcal{A}_{CS}}] < \mathbb{E}[C^{\mathcal{A}_{FP}}]$.

Scaling auctions in elastic environments. Theorem 1 implies that the scaling auction induces perverse incentives. Hence, by Proposition 1, $\varphi_{\mathcal{A}_S}$ is strictly decreasing and the project is awarded to the least efficient type, $\theta^{(I)}$. The boundary condition $U(\varphi_{\mathcal{A}_S}(\underline{\theta}), \underline{\theta} \mid \mathcal{A}_S) = 0$ and strict monotonicity imply $\mathbb{E}[C^{\mathcal{A}_S}] > \mathbb{E}[\text{EC}(\theta^{(I)})]$. By Lemma 4,

$$\mathbb{E}[C^{\mathcal{A}_S}] > \mathbb{E}[\text{EC}(\theta^{(I)})] \geq \mathbb{E}[\text{EC}(\theta^{(2)})] = \mathbb{E}[C^{\mathcal{A}_{FP}}].$$

Scaling auctions in inelastic environments. Theorem 1 implies that the scaling auction is well behaved. Write $\overline{Q}_t(\theta) := \mathbb{E}[q_t(\theta) + \xi_t]$. The expected utility of a risk-neutral winning seller can be written as $U(s, \theta \mid \mathcal{A}_S) = \frac{s}{w_t} \overline{Q}_t - \text{EC}(\theta)$ for some task t . Hence, $U_\theta(s, \theta \mid \mathcal{A}_S) = \frac{s}{w_t} \overline{Q}'_t(\theta) - M'(\theta)$. Since $\overline{Q}'_t(\theta) \geq 0$ and equilibrium scaling scores are positive, it follows that

$$U_\theta(\varphi_{\mathcal{A}_S}(\theta), \theta \mid \mathcal{A}_S) \geq -M'(\theta) = U_\theta(\varphi_{\mathcal{A}_{FP}}(\theta), \theta \mid \mathcal{A}_{FP}),$$

(with strict inequality whenever $\overline{Q}'_t(\theta) > 0$). Lemma 5 therefore implies $\mathbb{E}[C^{\mathcal{A}_S}] \leq \mathbb{E}[C^{\mathcal{A}_{FP}}]$, with strict inequality whenever the quantities are strictly increasing in type.

B Additional Empirical Results

Table 6: Summary statistics

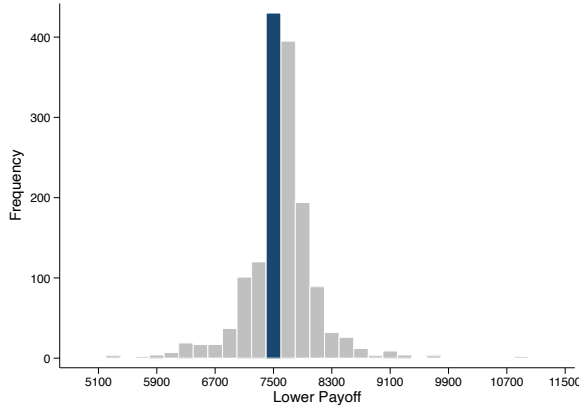
	Experiment 1	Experiment 2
CRT Score	1.74 (1.13)	1.72 (1.17)
Male	0.29 (0.46)	0.40 (0.49)
Age	23.54 (3.51)	21.94 (3.84)
English	0.09 (0.29)	0.30 (0.46)
Economics	0.25 (0.44)	0.42 (0.50)
Subjects	102	146

Notes: This table reports means for demographic variables separately by experiment. Standard deviations are in parentheses. CRT Score is the number of correct answers on a Cognitive Reflection Test, ranging from 0 to 3. Male, English, and Economics are indicator variables equal to one if the subject reports being male, speaking English as a first language, and majoring in economics, respectively.

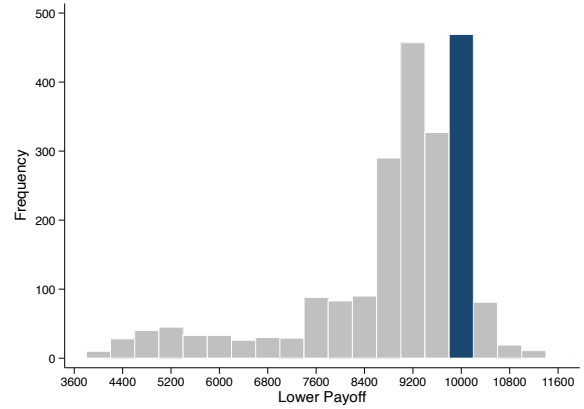
Table 7: Experiment 1: Efficiency and Procurement Costs (all rounds)

	(1)	(2)
	Total Costs	Procurement Costs
Scaling	506.6*** (36.9) [0.100]	601.4*** (51.4) [0.100]
Controls	Yes	Yes
Round Fixed Effects	Yes	Yes
R-Sqd	0.62	0.40
Observations	1530	1530

Notes: Standard errors clustered at the session level are in parentheses; two-sided randomization-inference p -values are in square brackets. Significance stars are based on the session-clustered t -tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

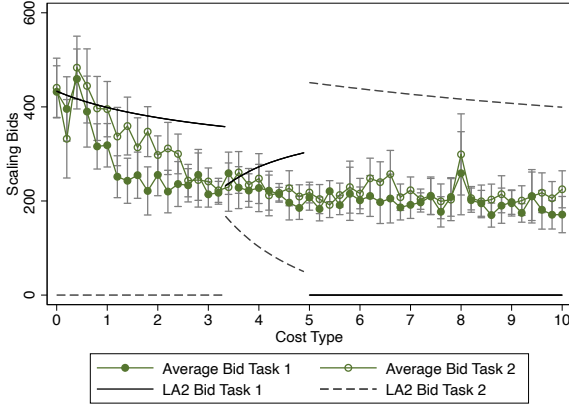


(a) Experiment 1

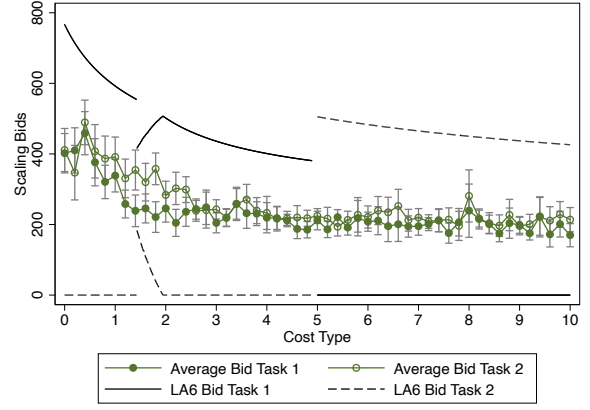


(b) Experiment 2

Figure 6: Histogram of Winners' Lower Payoff. This shows the distribution of winners' lower payoffs in the two experiments. The bin containing the outside option (the payoff one receives by losing the auction) is highlighted.



(a) $\lambda = 2$



(b) $\lambda = 6$

Figure 7: Average Bidding Behavior in Experiment 1 vs. Loss Aversion Predictions. Average bidding for cost type is plotted in green with a 95% confidence interval. The predicted bids under loss aversion for tasks 1 and 2 are presented with the solid and dashed lines, respectively.

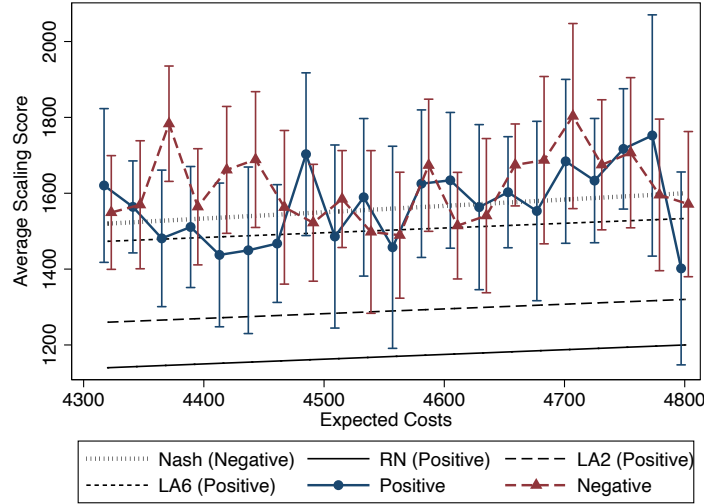


Figure 8: Scaling Scores in Experiment 2. The figure shows average scores along with the equilibrium predictions under risk neutrality, loss aversion with $\lambda = 2$, and loss aversion with $\lambda = 6$. Loss aversion does not change the theoretical prediction under negative correlation.

Table 8: Experiment 2 Outcomes (all rounds)

	(1) TC	(2) TC	(3) PC	(4) PC	(5) PD	(6) PD
Scaling	15.1* (7.71) [0.071]	20.7 (15.4) [0.433]	41.0 (379.5) [0.971]	763.7** (228.1) [0.445]	22.5 (558.1) [0.957]	1108.3** (430.8) [0.479]
Cost-Sharing	-10.5* (5.21) [0.057]	-18.4*** (1.96) [0.048]	1046.0*** (210.3) [0.100]	1161.1** (398.8) [0.524]	-342.5*** (33.4) [0.000]	-330.1*** (60.3) [0.233]
Scaling $\times$ Neg		-10.3 (15.2) [0.671]		-1336.8*** (287.6) [0.070]		-2008.6*** (433.5) [0.037]
Cost-Sharing $\times$ Neg		15.0** (6.21) [0.205]		-218.2 (415.3) [0.754]		-23.6 (67.2) [0.868]
t-test p-value: $S \times N = C \times N$		0.17		0.063		0.0027
RI p-value: $S \times N = C \times N$		0.29		0.22		0.013
Controls	Yes	Yes	Yes	Yes	Yes	Yes
R-Sqd	0.53	0.53	0.18	0.20	0.20	0.34
Observations	2190	2190	2190	2190	2190	2190

Notes: “TC” stands for “total costs,” “PC” stands for “procurement costs,” and “PD” stands for “payoff difference.” Standard errors clustered at the session level are in parentheses; two-sided randomization-inference p -values are in square brackets. Significance stars are based on the session-clustered t -tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 9: Experiment 2 Outcomes (Contingent Auctions Only)

	(1) TC	(2) TC	(3) PC	(4) PC	(5) PD	(6) PD
Scaling	13.1*** (2.45) [0.005]	11.1* (4.91) [0.187]	99.0 (339.3) [0.763]	788.1 (449.0) [0.259]	387.2 (448.4) [0.517]	1538.4*** (434.2) [0.095]
Neg	7.46** (2.70) [0.105]	5.51*** (1.06) [0.152]	-594.9 (347.4) [0.206]	59.3 (71.2) [0.759]	-1133.5** (457.6) [0.063]	-40.6 (72.9) [0.784]
Scaling $\times$ Neg		3.83 (4.85) [0.632]		-1289.3** (506.3) [0.010]		-2153.8*** (439.1) [0.013]
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Round Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
R-Sqd	0.53	0.53	0.095	0.18	0.21	0.37
Observations	949	949	949	949	949	949

Notes: “TC” stands for “total costs,” “PC” stands for “procurement costs,” and “PD” stands for “payoff difference.” Standard errors clustered at the session level are in parentheses; two-sided randomization-inference p -values are in square brackets. Significance stars are based on the session-clustered t -tests: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 10: Smoothed Bidding Counterfactuals

Format	Correlation	Winner expected total cost		Government procurement cost	
		Actual	Smoothed	Actual	Smoothed
Fixed-Payment	Positive	4536.9	4527.3	4179.0	4780.1
Fixed-Payment	Negative	4539.7	4514.3	4379.2	4965.0
Cost-Sharing	Positive	4541.2	4505.5	4704.6	4818.8
Cost-Sharing	Negative	4527.4	4481.6	4744.5	4853.2
Scaling	Positive	4532.2	4498.6	5469.7	6180.9
Scaling	Negative	4552.7	4536.8	4253.3	4705.0

Notes: This table presents the actual and smoothed auction outcomes for Experiment 2. Actual outcomes are the realized values based on subjects’ bids. Smoothed outcomes are obtained first by replacing each bidder’s submitted bid with the average bid conditional on the bidder’s signal then by computing the auction’s results with these new bids. Bidding averages are computed separately for each auction format and correlation structure. For scaling auctions, the per-task bids are smoothed separately. When bids are tied under smoothed bids, outcomes are averaged across the tied bidders. Across all formats and correlations, smoothing decreases total costs and increases procurement costs.

C Verification of Equilibria in Examples

In this section, we verify the equilibria from the examples in Section 3.3.

Fixed-Payment Auctions The proposed equilibrium bid is $\varphi_{FP}(\theta) = \frac{29}{2} + \frac{1}{2}\theta$. We verify that a seller of type θ would not want to imitate the bid of any other type $\hat{\theta}$ who bids $\frac{29}{2} + \frac{1}{2}\hat{\theta}$ and wins with chance $1 - \hat{\theta}$. The seller solves

$$\max_{\hat{\theta}} (1 - \hat{\theta}) \left[\frac{29}{2} + \frac{1}{2}\hat{\theta} - 14 - \theta \right].$$

Differentiating the objective function with respect to $\hat{\theta}$ gives $\theta - \hat{\theta}$, which is positive for $\hat{\theta} < \theta$ and negative for $\hat{\theta} > \theta$, so the seller maximizes their payoffs by setting $\hat{\theta} = \theta$, verifying the equilibrium.

The density of the winner's type in equilibrium is $2 - 2\theta$, so expected procurement costs are

$$\int_0^1 \left(\frac{29}{2} + \frac{1}{2}\theta \right) (2 - 2\theta) d\theta = \frac{44}{3}.$$

Cost-Sharing Auctions under Fixed Costs Heterogeneity The proposed equilibrium bid is $\varphi_{CS} = \frac{5}{8} + \frac{1}{8}\theta$. We verify that a seller of type θ would not want to imitate the bid of any other type $\hat{\theta}$ who bids $\frac{5}{8} + \frac{1}{8}\hat{\theta}$ and wins with chance $1 - \hat{\theta}$. The seller solves

$$\max_{\hat{\theta}} (1 - \hat{\theta}) \left[12 + \left(\frac{5}{8} + \frac{1}{8}\hat{\theta} \right) (4) - 14 - \theta \right]$$

Differentiating the objective function with respect to $\hat{\theta}$ gives $\theta - \hat{\theta}$, which is positive for $\hat{\theta} < \theta$ and negative for $\hat{\theta} > \theta$, so the seller maximizes their payoffs by setting $\hat{\theta} = \theta$, verifying the

equilibrium.

The density of the winner's type in equilibrium is $2 - 2\theta$, so expected procurement costs are

$$\int_0^1 \left(12 + \left(\frac{5}{8} + \frac{1}{8}\theta \right) (4) \right) (2 - 2\theta) d\theta = \frac{44}{3}.$$

Cost-Sharing Auctions under Marginal Costs or Input Quantities Heterogeneity

The proposed equilibrium bid is $\varphi_{CS}(\theta) = 1 + \frac{2}{1-\theta} \ln \left(\frac{4+\theta}{5} \right)$ for $\theta < 1$ and $\varphi_{CS}(1) = \frac{3}{5}$. We verify that the seller of type θ would not want to imitate the bid of any other type $\hat{\theta}$ who bids $1 + \frac{2}{1-\hat{\theta}} \ln \left(\frac{4+\hat{\theta}}{5} \right)$ and wins with chance $1 - \hat{\theta}$. The seller solves

$$\max_{\hat{\theta}} \left(1 - \hat{\theta} \right) \left[12 + \left(1 + \frac{2}{1-\hat{\theta}} \ln \left(\frac{4+\hat{\theta}}{5} \right) \right) (4 + \theta) - 14 - \theta \right].$$

Differentiating the objective function with respect to $\hat{\theta}$ gives $\frac{2(\theta-\hat{\theta})}{\hat{\theta}+4}$, which is positive for $\hat{\theta} < \theta$ and negative for $\hat{\theta} > \theta$, so the seller maximizes their payoffs by setting $\hat{\theta} = \theta$, verifying the equilibrium.

The density of the winner's type in equilibrium is $2 - 2\theta$, so expected procurement costs are

$$\int_0^1 \left(12 + \left(1 + \frac{2}{1-\theta} \ln \left(\frac{4+\theta}{5} \right) \right) (4 + \theta) \right) (2 - 2\theta) d\theta = \frac{22}{3} + 32 \ln \left(\frac{5}{4} \right).$$

Scaling Auctions under Fixed Costs Heterogeneity The two tasks have the same expected quantities and weights, so for a fixed score, any feasible combination of per-task bids gives the same expected payoffs. Thus, in our calculations we assume that the score is

split equally between the two tasks. The proposed equilibrium score is $\varphi_S(\theta) = \frac{29}{4} + \frac{1}{4}\theta$. We verify that the seller of type θ would not want to imitate the score of any other type $\hat{\theta}$ who bids $\frac{29}{4} + \frac{1}{4}\hat{\theta}$ and wins with chance $1 - \hat{\theta}$. The seller solves

$$\max_{\hat{\theta}} \left(1 - \hat{\theta}\right) \left[(2)(2) \left(\frac{29}{8} + \frac{1}{8}\hat{\theta} \right) - 14 - \theta \right].$$

Differentiating the objective function with respect to $\hat{\theta}$ gives $\theta - \hat{\theta}$, which is positive for $\hat{\theta} < \theta$ and negative for $\hat{\theta} > \theta$, so the seller maximizes their payoffs by setting $\hat{\theta} = \theta$, verifying the equilibrium.

The density of the winner's type in equilibrium is $2 - 2\theta$, so expected procurement costs are

$$\int_0^1 \left((2)(2) \left(\frac{29}{8} + \frac{1}{8}\theta \right) \right) (2 - 2\theta) d\theta = \frac{44}{3}.$$

Scaling Auctions under Input Quantities Heterogeneity The two tasks have the same expected quantities and weights, so for a fixed score, any feasible combination of per-task bids gives the same expected payoffs. Thus, in our calculations we assume that the score is split equally between the two tasks. The proposed equilibrium score is $\varphi_S(\theta) = 2 + \frac{20}{\theta} \ln \left(\frac{4+\theta}{4} \right)$ for $\theta > 0$ and $\varphi_S(0) = 7$. We verify that the seller of type θ would not want to imitate the score of any other type $\hat{\theta}$ who bids $2 + \frac{20}{\hat{\theta}} \ln \left(\frac{4+\hat{\theta}}{4} \right)$ and wins with chance $\hat{\theta}$. The seller solves

$$\max_{\hat{\theta}} \hat{\theta} \left[2 \left(1 + \frac{20}{2\hat{\theta}} \ln \left(\frac{4+\hat{\theta}}{4} \right) \right) \left(2 + \frac{1}{2}\theta \right) - 14 - \theta \right]$$

Differentiating the objective function with respect to $\hat{\theta}$ gives $\frac{10(\theta-\hat{\theta})}{\theta+4}$, which is positive for $\hat{\theta} < \theta$ and negative for $\hat{\theta} > \theta$, so the seller maximizes their payoffs by setting $\hat{\theta} = \theta$, verifying the equilibrium. The density of the winner's type in equilibrium is 2θ , so expected procurement costs are

$$\int_0^1 2 \left(1 + \frac{20}{2\theta} \ln \left(\frac{4+\theta}{4} \right) \right) \left(2 + \frac{1}{2}\theta \right) (2\theta) d\theta = 250 \ln \left(\frac{5}{4} \right) - \frac{121}{3}$$

Scaling Auctions under Marginal Costs (+) Heterogeneity The two tasks have the same expected quantities and weights, so for a fixed score, any feasible combination of per-task bids gives the same expected payoffs. Thus, in our calculations we assume that the score is split equally between the two tasks. The proposed equilibrium score is $\varphi_S(\theta) = \frac{29}{6} + \frac{1}{6}\theta$. We verify that the seller of type θ would not want to imitate the score of any other type $\hat{\theta}$ who bids $\frac{29}{6} + \frac{1}{6}\hat{\theta}$ and wins with chance $1 - \hat{\theta}$. The seller solves

$$\max_{\hat{\theta}} \left(1 - \hat{\theta} \right) \left[2 \left(\frac{29}{12} + \frac{1}{12}\hat{\theta} \right) (3) - 14 - \theta \right].$$

Differentiating the objective function with respect to $\hat{\theta}$ gives $\theta - \hat{\theta}$, which is positive for $\hat{\theta} < \theta$ and negative for $\hat{\theta} > \theta$, so the seller maximizes their payoffs by setting $\hat{\theta} = \theta$, verifying the equilibrium.

The density of the winner's type in equilibrium is $2 - 2\theta$, so expected procurement costs are

$$\int_0^1 2 \left(\frac{29}{12} + \frac{1}{12}\hat{\theta} \right) (3) (2 - 2\theta) d\theta = \frac{44}{3}$$

Scaling Auctions under Marginal Costs (–) Heterogeneity The two tasks have the same expected quantities and weights, so for a fixed score, any feasible combination of per-task bids gives the same expected payoffs. Thus, in our calculations we assume that the score is split equally between the two tasks. The proposed equilibrium score is $\varphi_S(\theta) = \frac{29}{6} + \frac{1}{6}\theta$. We verify that the seller of type θ would not want to imitate the score of any other type $\hat{\theta}$ who bids $\frac{29}{6} + \frac{1}{6}\hat{\theta}$ and wins with chance $1 - \hat{\theta}$. The seller solves

$$\max_{\hat{\theta}} \left(1 - \hat{\theta}\right) \left[2 \left(\frac{29}{12} + \frac{1}{12}\hat{\theta}\right) (3) - 14 - \theta\right].$$

It is immediate that this is equivalent to the above case of Scaling Auctions under Marginal Costs (+) Heterogeneity, and thus the same results follow.

D Screenshots and Instructions

D.1 Experiment 1

Below, we include screenshots from Experiment 1.

Introduction

PLEASE READ CAREFULLY AND DO NOT PRESS NEXT UNTIL INSTRUCTED TO DO SO.

Thank you for participating in this study. This study is about decision-making. It should take about 90-120 minutes, and you will be paid based on your earnings from the experiment. The money you earn will be paid either in cash at the end of the study or electronically within a few days of the end of the study.

Please do not use any electronic devices or talk with other participants during this study.

There will be no deception in this study. Every decision you make will be carried out exactly as it is described in the instructions. Anything else would violate the human ethics protocol under which we run the study (UQ Human Research Ethics Approval 2025/HE001937).

In the study you will make decisions that will affect the amount of money you earn. The study will consist of games that you will play with other randomly selected players. The players that you are paired with in a match are selected independently of who you play with in any other match.

Please pay close attention to the instructions on the next page. After you read these instructions, there will be a short quiz on the instructions. You will receive \$1 for each question you answer correctly.

If you have questions at any point, please raise your hand and we will answer your questions privately.

Next

Figure 9: Introduction

Instructions

In this study you will participate in 30 rounds of "reverse" auctions: the government needs someone to complete a project, and participants in the auction make bids to determine who will complete the project and how much they will be paid.

You will always see numbers in **points**. At the end, one round is picked at random to pay you at 150 points = \$1. You start each round with an **endowment of 7,500 points**.

Every round, you're matched with **one other bidder**, and you will be randomly rematched in the next round. The auctions will follow a fixed set of rules, which we describe on the next page.

About the Project

The project requires two different **inputs**. You can think of these as two kinds of work or materials. What matters is how many *units* of each input are needed.

- **Input 1:** baseline is either 0 units (low) or 6 units (high).
- **Input 2:** baseline is either 0 units (low) or 2 units (high).

At the start of each round, there is a 50–50 chance the project will need the **low** baseline quantities or the **high** baseline quantities. The two inputs always move together: if Input 1 is low, Input 2 is also low; if Input 1 is high, Input 2 is also high. However, you will never know whether you will end up needing the low or the high baseline quantities.

On top of these baselines, **each bidder gets their own personal add-on** for both inputs. This add-on is a random number between 0 and 10, drawn separately for each person. Each number between 0 and 10 (in increments of 0.2) is equally likely to be chosen. Your actual required quantity of each input is therefore:

- Baseline quantity (low or high with equal chance), plus
- Your personal random add-on (unique to you).

You will see *your own input quantities* on screen. Your opponent will see theirs, which will be different because of the personal add-ons. You never see your opponent's quantities.

If you win the project, you must pay both a **fixed cost** (which does not depend on the input quantities) and **variable costs** (which do depend on them). The fixed cost is **500 points**. Each unit of Input 1 costs **150 points**, and each unit of Input 2 costs **100 points**. Your variable costs are the total of these unit costs multiplied by your actual input quantities.

Next

Figure 10: Instructions Page 1

Instructions

Auction Rules

In these auctions, you will use a slider to choose the **percent of variable costs you request to be covered**. This is your **bid**.

- **Lower bids** mean the government covers a *smaller* share (you cover more).
- The bidder who asks for the **smallest share** to be covered wins.

The winner will receive a fixed upfront amount of **1800 points**.

What happens if you win?

Your final points for the round are:

- **Your endowment** (7,500), plus the **upfront amount** (1800),
- minus **your share of the project's variable costs** (based on the inputs actually used),
- minus the **fixed cost** of 500 points.

The cost per unit is 150 for Input 1 and 100 for Input 2. The project's variable cost depends on how many units of each input are used in that round (which can be low or high, plus your personal add-ons). "Your share of the cost" depends on your bid. For example, if you bid 40%, then 40% of the variable costs will be covered and you pay the remaining 60%.

What happens if you lose?

You simply keep your endowment (7,500) for that round (no gains or losses from the project).

After each round

- The loser sees only their own result.
- The winner sees their own result and a reminder of what the loser's payoff would have been.

Bidding & Interface

- You will place bids using **sliders** on the screen.
- You will see **your own possible input quantities** (including your personal add-ons) and the possible net payoffs you would receive for winning the auction.
- You will **not** see your opponent's quantities or bids during the round.

Payment

- At the end of the experiment, one round will be chosen at random to be the round that counts. Your payment will be determined by the number of points you earned in that round

On the next page, you will find a few examples to help you better understand the rules. Please go through them at your own pace. After you have reviewed the examples, click Next again to begin the quiz.

Next

Figure 11: Instructions Page 2 (Cost-Sharing Treatment)

Instructions

Auction Rules

In these auctions, you will use a slider to submit two **unit prices** that you want to be paid for the inputs: one price per unit of Input 1 and one price per unit of Input 2. The government uses two fixed **weights**, $w_1 = 8.0$ and $w_2 = 6.0$, to make a **score**: $\text{Score} = (\text{your Input 1 price}) \times w_1 + (\text{your Input 2 price}) \times w_2$.

- **Lower unit prices** mean the government pays a *smaller* amount per unit (you receive less).
- The bidder with the **lowest score** wins.

What happens if you win?

Your final points for the round are:

- **Your endowment** (7,500),
- plus, for each input: **(your unit price – cost per unit) $\times$ actual units**,
- minus the **fixed cost** of 500 points.

The cost per unit is 150 for Input 1 and 100 for Input 2. If your price is above cost, you gain on that input; if it's below cost, you lose on that input. The actual units can be low or high, plus your personal add-ons.

What happens if you lose?

You simply keep your endowment (7,500) for that round (no gains or losses from the project).

After each round

- The loser sees only their own result.
- The winner sees their own result and a reminder of what the loser's payoff would have been.

Bidding & Interface

- You will place bids using **sliders** on the screen.
- You will see **your own possible input quantities** (including your personal add-ons) and the possible net payoffs you would receive for winning the auction.
- You will **not** see your opponent's quantities or bids during the round.

Payment

- At the end of the experiment, one round will be chosen at random to be the round that counts. Your payment will be determined by the number of points you earned in that round

On the next page, you will find a few examples to help you better understand the rules. Please go through them at your own pace. After you have reviewed the examples, click Next again to begin the quiz.

Next

Figure 12: Instructions Page 2 (Scaling Treatment)

Examples

PLEASE READ THE EXAMPLES CAREFULLY AND CLICK NEXT WHEN YOU ARE READY TO BEGIN THE QUIZ.

Example 1

- Ann and Bob are both participating in the auction.
- Ann finds out that her personal input add-on is 7.4. That means that:
 - If the baseline requirement is low (50% chance) she will require 7.4 units of Input 1 and 7.4 units of Input 2.
 - If the baseline requirement is high (50% chance) she will require 13.4 units of Input 1 and 9.4 units of Input 2.
- Ann bids 40%, meaning that if she wins the government will pay 40% of the variable costs and Anne will pay 60% of the variable costs. Bob bids 53%, so Ann wins.
- Because she won the auction, Ann's payoffs are
 - 50% chance of **7690** = $7500 + 1800 - 0.60 \times (7.4 \times 150 + 7.4 \times 100) - 500$
 - 50% chance of **7030** = $7500 + 1800 - 0.60 \times (13.4 \times 150 + 9.4 \times 100) - 500$
- Because he did not win the auction, Bob receives 7500 with certainty.

Example 2

- Carlos and Daniella are both participating in the auction.
- Carlos finds out that his personal input add-on is 1.6. That means that:
 - If the baseline requirement is low (50% chance) he will require 1.6 units of Input 1 and 1.6 units of Input 2.
 - If the baseline requirement is high (50% chance) he will require 7.6 units of Input 1 and 3.6 units of Input 2.
- Carlos bids 32%, meaning that if he wins the government will pay 32% of the variable costs and Carlos will pay 68% of the variable costs. Daniella bids 78%, so Carlos wins.
- Because he won the auction, Carlos' payoffs are
 - 50% chance of **8560** = $7500 + 1800 - 0.68 \times (1.6 \times 150 + 1.6 \times 100) - 500$
 - 50% chance of **7780** = $7500 + 1800 - 0.68 \times (7.6 \times 150 + 3.6 \times 100) - 500$
- Because she did not win the auction, Daniella receives 7500 with certainty.

When you are ready, click Next again to begin the quiz.

Next

Figure 13: Examples (Cost-Sharing Treatment)

Examples

PLEASE READ THE EXAMPLES CAREFULLY AND CLICK NEXT WHEN YOU ARE READY TO BEGIN THE QUIZ.

Example 1

- Ann and Bob are both participating in the auction.
- Ann finds out that her personal input add-on is 7.4. That means that:
 - If the baseline requirement is low (50% chance) she will require 7.4 units of Input 1 and 7.4 units of Input 2.
 - If the baseline requirement is high (50% chance) she will require 13.4 units of Input 1 and 9.4 units of Input 2.
- Ann bids 250 for Input 1 and 80 for Input 2, meaning that her score is $8 \times 250 + 6 \times 80 = 2480$. Bob's score was 2900, so Ann wins.
- Because she won the auction, Ann's payoffs are
 - 50% chance of **7592** = $7500 + 7.4 \times (250 - 150) + 7.4 \times (80 - 100) - 500$
 - 50% chance of **8152** = $7500 + 13.4 \times (250 - 150) + 9.4 \times (80 - 100) - 500$
- Because he did not win the auction, Bob receives 7500 with certainty.

Example 2

- Carlos and Daniella are both participating in the auction.
- Carlos finds out that his personal input add-on is 1.6. That means that:
 - If the baseline requirement is low (50% chance) he will require 1.6 units of Input 1 and 1.6 units of Input 2.
 - If the baseline requirement is high (50% chance) he will require 7.6 units of Input 1 and 3.6 units of Input 2.
- Carlos bids 200 for Input 1 and 300 for Input 2, meaning that his score is $8 \times 200 + 6 \times 300 = 3400$. Daniella's score was 4500, so Carlos wins.
- Because he won the auction, Carlos' payoffs are
 - 50% chance of **7400** = $7500 + 1.6 \times (200 - 150) + 1.6 \times (300 - 100) - 500$
 - 50% chance of **8100** = $7500 + 7.6 \times (200 - 150) + 3.6 \times (300 - 100) - 500$
- Because she did not win the auction, Daniella receives 7500 with certainty.

When you are ready, click Next again to begin the quiz.

Next

Figure 14: Examples (Scaling Treatment)

Quiz

You will now be given a series of questions to check your understanding of the instructions and examples. You will be paid \$1 for each answer you get correct.

At the end of the experiment, how is your final payment determined?

- ☐ By summing your total points across all rounds
- ☐ By averaging your points across all rounds
- ☐ By selecting one round at random and paying based on its points
- ☐ By paying based on a round you won the auction

Which of the following is true about the project's input requirements?

- ☐ Only one input is used in each project
- ☐ Inputs 1 and 2 move together: both low or both high
- ☐ Each bidder sees both their own and their opponent's add-ons
- ☐ The baseline for Input 1 is always zero

What happens if you lose an auction round?

- ☐ You pay only the fixed cost
- ☐ You get your endowment only
- ☐ You receive the government's payment
- ☐ You must skip the next round

How do you place your bid?

- ☐ Submit unit prices for each input
- ☐ Choose the percent of variable costs to be covered
- ☐ State a lump-sum payment request
- ☐ Indicate how many inputs you will supply

How is the winner determined?

- ☐ Lowest requested share to be covered
- ☐ Bid is closest to the average
- ☐ Largest endowment remaining
- ☐ Smallest add-on

When you believe you have answered all questions correctly, press Next to check your answers.

Next

Figure 15: Quiz (Cost-Sharing Treatment)

Quiz

You will now be given a series of questions to check your understanding of the instructions and examples. You will be paid \$1 for each answer you get correct.

At the end of the experiment, how is your final payment determined?

- ☐ By summing your total points across all rounds
- ☐ By averaging your points across all rounds
- ☐ By selecting one round at random and paying based on its points
- ☐ By paying based on a round you won the auction

Which of the following is true about the project's input requirements?

- ☐ Only one input is used in each project
- ☐ Inputs 1 and 2 move together: both low or both high
- ☐ Each bidder sees both their own and their opponent's add-ons
- ☐ The baseline for Input 1 is always zero

What happens if you lose an auction round?

- ☐ You pay only the fixed cost
- ☐ You get your endowment only
- ☐ You receive the government's payment
- ☐ You must skip the next round

How do you place your bid?

- ☐ Choose the percent of variable costs to be covered
- ☐ Submit unit prices for each input
- ☐ State a lump-sum payment request
- ☐ Indicate how many inputs you will supply

How is the winner determined?

- ☐ Lowest weighted score from unit prices
- ☐ Highest unit price for Input 1
- ☐ Largest endowment remaining
- ☐ Smallest add-on

When you believe you have answered all questions correctly, press Next to check your answers.

Next

Figure 16: Quiz (Scaling Treatment)

Quiz Answers

The answers for the quiz are given below. Please review the answers and note any mistakes you have made.

Question 1: At the end of the experiment, how is your final payment determined?

Correct Answer: By selecting one round at random and paying based on its points

Your Answer: By selecting one round at random and paying based on its points

Question 2: Which of the following is true about the project's input requirements?

Correct Answer: Inputs 1 and 2 move together: both low or both high

Your Answer: Inputs 1 and 2 move together: both low or both high

Question 3: What happens if you lose an auction round?

Correct Answer: You get your endowment only

Your Answer: You get your endowment only

Question 4: How do you place your bid?

Correct Answer: Choose the percent of variable costs to be covered

Your Answer: Choose the percent of variable costs to be covered

Question 5: How is the winner determined?

Correct Answer: Lowest requested share to be covered

Your Answer: Lowest requested share to be covered

You earned \$5.0 from your correct answers. Please review any questions you answered incorrectly. When you are ready to begin the first round, click the next button.

Next

Figure 17: Quiz Answers (Cost-Sharing Treatment)

Quiz Answers

The answers for the quiz are given below. Please review the answers and note any mistakes you have made.

Question 1: At the end of the experiment, how is your final payment determined?

Correct Answer: By selecting one round at random and paying based on its points

Your Answer: By paying based on a round you won the auction

Question 2: Which of the following is true about the project's input requirements?

Correct Answer: Inputs 1 and 2 move together: both low or both high

Your Answer: Inputs 1 and 2 move together: both low or both high

Question 3: What happens if you lose an auction round?

Correct Answer: You get your endowment only

Your Answer: You get your endowment only

Question 4: How do you place your bid?

Correct Answer: Submit unit prices for each input

Your Answer: Indicate how many inputs you will supply

Question 5: How is the winner determined?

Correct Answer: Lowest weighted score from unit prices

Your Answer: Lowest weighted score from unit prices

You earned \$3.0 from your correct answers. Please review any questions you answered incorrectly. When you are ready to begin the first round, click the next button.

Next

Figure 18: Quiz Answers (Scaling Treatment)

Auction Round 1

Your personal input add-on for this round is 1.6. That means that

- If the baseline requirement is low, you will require 1.6 units of Input 1 and 1.6 units of Input 2.
- If the baseline requirement is high, you will require 7.6 units of Input 1 and 3.6 of Input 2.

Use the slider to choose the **percent of variable costs you request to be covered**. The bidder asking for the **smallest share** wins.

If you win: $7,500 + \text{fixed transfer (1800)} - \text{your share of variable costs} - 500$. Variable costs are 150 per unit of Input 1 and 100 per unit of Input 2.

If you lose: you keep 7,500.

You see only your own input quantities. Your opponent's are hidden.

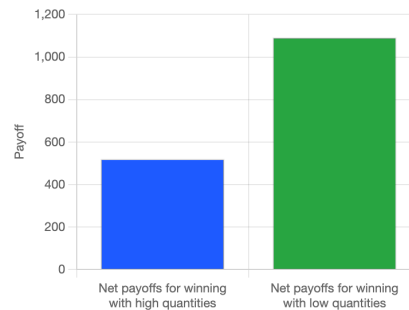
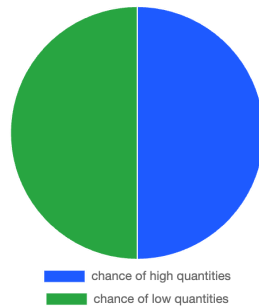
0

100

Bid (%): 48

Click "Next" when you're ready to submit your bid.

Next



Key Reminders

- Only **one randomly selected round** will be paid in cash.
- Your endowment is **7,500 points**.
- Conversion rate: **150 points = \$1**.
- Fixed cost if you win: **500 points**.
- Per-unit variable costs: **Input 1 = 150, Input 2 = 100**.
- Baselines: Input 1 **Low = 0, High = 6**; Input 2 **Low = 0, High = 2**.
- Personal add-ons: each input gets **+ a random 0 to 10 units** unique to you.

Figure 19: Bidding Page (Cost-Sharing Treatment)

Auction Round 1

Your personal input add-on for this round is 1.2. That means that

- If the baseline requirement is low, you will require 1.2 units of Input 1 and 1.2 units of Input 2.
- If the baseline requirement is high, you will require 7.2 units of Input 1 and 3.2 of Input 2.

Set your **bid per unit** for Input 1 and Input 2 using the sliders. These bids will be combined into a score using fixed weights. The **lowest score wins**.

If you win: $7,500 + (\text{your bid} - \text{cost}) \times \text{units} - 500$. Variable costs are 150 per unit of Input 1 and 100 per unit of Input 2.

If you lose: you keep 7,500.

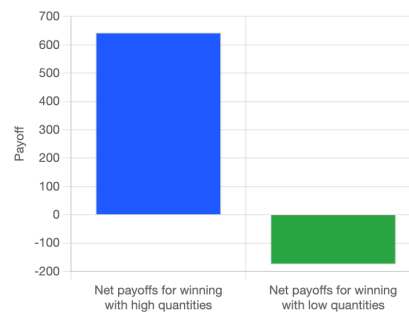
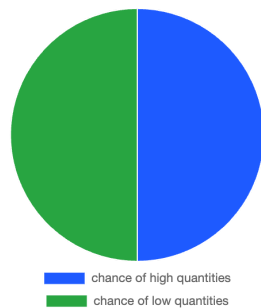
You see only your own input quantities. Your opponent's are hidden.

0		600
Input 1 Bid: 220		
0		600
Input 2 Bid: 300		

Your score for this auction is $8 \times 220 + 6 \times 300 = 3560$. Remember that the bidder with the lowest score wins.

Click "Next" when you're ready to submit your bid.

Next



Key Reminders

- Only **one randomly selected round** will be paid in cash.
- Your endowment is **7,500 points**.
- Conversion rate: **150 points = \$1**.
- Fixed cost if you win: **500 points**.
- Per-unit variable costs: **Input 1 = 150, Input 2 = 100**.
- Baselines: Input 1 **Low = 0, High = 6**; Input 2 **Low = 0, High = 2**.
- Personal add-ons: each input gets **+ a random 0 to 10 units** unique to you.

Figure 20: Bidding Page (Scaling Treatment)

Round 1 Results

Congratulations! You won the auction.

Your bid was 24% and the other bid was 84%.

Your payoff for this round will be $7500 + 1800 - (1-0.24)((150 \times 7.4) + (100 \times 3.4)) - 500 = \mathbf{7698.0}$ with one-half chance and $7500 + 1800 - (1-0.24)((150 \times 1.4) + (100 \times 1.4)) - 500 = \mathbf{8534.0}$ with one-half chance.

The bidder that did not win will receive 7500 with certainty.

Next

Figure 21: Winner's Results (Cost-Sharing Treatment)

Round 1 Results

Congratulations! You won the auction.

Your score was 3200.0 and the other bidder's score was 7800.0.

Your payoff for this round will be $7500 + (160 - 150) \times 12.4 + (320 - 100) \times 8.4 - 500 = \mathbf{8972.0}$ with one-half chance and $7500 + (160 - 150) \times 6.4 + (320 - 100) \times 6.4 - 500 = \mathbf{8472.0}$ with one-half chance.

The bidder that did not win will receive 7500 with certainty.

Next

Figure 22: Winner's Results (Scaling Treatment)

Round 1 Results

You did not win this round.

Your bid was 84% and the other bid was 24%.

Your payoff for this round will be 7500.

Next

Figure 23: Loser's Results (Cost-Sharing Treatment)

Round 1 Results

You did not win this round.

Your score was 7800.0 and the other bidder's score was 3200.0.

Your payoff for this round will be 7500.

Next

Figure 24: Loser's Results (Scaling Treatment)

Survey

Please answer the following questions.

What is your age?

What gender do you identify with the most?

- ☐ Female
- ☐ Male
- ☐ Other/Prefer Not to Say

Is English your first language?

- ☐ Yes
- ☐ No

Are you completing or have you completed an economics degree?

- ☐ Yes
- ☐ No

Next

Figure 25: Demographic Survey

Survey

Please answer the following questions.

A bat and a ball cost 22 dollars in total. The bat costs 20 dollars more than the ball. How many dollars does the ball cost?

"If it takes 5 machines 5 minutes to make 5 widgets, how many minutes would it take 100 machines to make 100 widgets?"

In a lake, there is a patch of lily pads. Every day, the patch doubles in size. If it takes 48 days for the patch to cover the entire lake, how many days would it take for the patch to cover half of the lake?

Next

Figure 26: Cognitive Reflection Test

Survey

Please answer the following questions:

The instructions of this study were easy to understand.

☐ Strongly disagree ☐ Disagree ☐ Neither agree nor disagree ☐ Agree ☐ Strongly agree

What, if anything, were you confused about in the study?

I knew how to make the decisions that were best for me in the experiment.

☐ Strongly disagree ☐ Disagree ☐ Neither agree nor disagree ☐ Agree ☐ Strongly agree

How did you make decisions in the study?

What do you think this study was about?

Next

Figure 27: Experiment Feedback

D.2 Experiment 2

Below, we include screenshots from Experiment 2.

Introduction

PLEASE READ CAREFULLY AND DO NOT PRESS NEXT UNTIL INSTRUCTED TO DO SO.

Thank you for participating in this study. This study is about decision-making. It should take about 90-120 minutes, and you will be paid based on your earnings from the experiment. The money you earn will be paid either in cash at the end of the study or electronically within a few days of the end of the study.

Please do not use any electronic devices or talk with other participants during this study.

There will be no deception in this study. Every decision you make will be carried out exactly as it is described in the instructions. Anything else would violate the human ethics protocol under which we run the study (UQ Human Research Ethics Approval 2025/HE001937).

In the study you will make decisions that will affect the amount of money you earn. The study will consist of games that you will play with other randomly selected players. The players that you are paired with in a match are selected independently of who you play with in any other match.

Please pay close attention to the instructions on the next page. After you read these instructions, there will be a short quiz on the instructions. You will receive \$1 for each question you answer correctly.

If you have questions at any point, please raise your hand and we will answer your questions privately.

Next

Figure 28: Introduction

Instructions

PLEASE READ CAREFULLY AND DO NOT PRESS NEXT UNTIL INSTRUCTED TO DO SO.

In this study you will participate in 30 rounds of "reverse" auctions: the government needs someone to complete a project, and participants in the auction make bids to determine who will complete the project and how much they will be paid.

You will always see numbers in **points**. At the end, one round is picked at random to pay you at 200 points = \$1. You start each round with an **endowment of 10000 points**.

Every round, you're matched with **one other bidder**, and you will be randomly rematched in the next round. You will participate in two types of auctions, which we describe on the next two pages.

About the Project

The project requires two different **inputs**. You can think of these as two kinds of work or materials. What matters is how many *units* of each input are needed.

- **Input 1**: either 2 units (Set A) or 6 units (Set B).
- **Input 2**: either 2 units (Set A) or 6 units (Set B).

At the start of each round, there is a 50–50 chance the project will require **Set A** quantities or **Set B** quantities. You will not know which set applies when you bid.

If you win the project, you must pay two types of costs: a **fixed cost** (which does not depend on input quantities) and **variable costs** (which do depend on input quantities). The fixed cost is **2400 points**.

In each round, your per-unit input costs are **randomly drawn** and **can differ across participants**.

The per-unit cost of Input 1 is equally likely to take any value between 320 and 400 (in steps of 4). The per-unit cost of Input 2 is equally likely to take any value between 160 and 200 (in steps of 2). **These two costs are perfectly linked: in every round, Input 1's unit cost is exactly two times Input 2's unit cost.**

Your total variable cost is: (unit cost of Input 1 × your quantity of Input 1) + (unit cost of Input 2 × your quantity of Input 2).

Bidding & Interface

- You will place bids using **sliders** on the screen.
- You will see **your own per-unit costs** and the possible net payoffs you would receive for winning the auction.
- You will **not** see your opponent's per-unit costs.

Feedback

After each round:

- The loser sees only their own result.
- The winner sees their own result and a reminder of what the loser's payoff is.

Payment

- At the end of the experiment, one round will be chosen at random to be the round that counts. Your payment will be determined by the number of points you earned in that round.

Next

Figure 29: Positive Correlation Instructions

Instructions

PLEASE READ CAREFULLY AND DO NOT PRESS NEXT UNTIL INSTRUCTED TO DO SO.

In this study you will participate in 30 rounds of "reverse" auctions: the government needs someone to complete a project, and participants in the auction make bids to determine who will complete the project and how much they will be paid.

You will always see numbers in **points**. At the end, one round is picked at random to pay you at 200 points = \$1. You start each round with an **endowment of 10000 points**.

Every round, you're matched with **one other bidder**, and you will be randomly rematched in the next round. You will participate in two types of auctions, which we describe on the next two pages.

About the Project

The project requires two different **inputs**. You can think of these as two kinds of work or materials. What matters is how many *units* of each input are needed.

- **Input 1**: either 1 unit (Set A) or 5 units (Set B).
- **Input 2**: either 5 units (Set A) or 1 unit (Set B).

At the start of each round, there is a 50–50 chance the project will require **Set A** quantities or **Set B** quantities. You will not know which set applies when you bid.

If you win the project, you must pay two types of costs: a **fixed cost** (which does not depend on input quantities) and **variable costs** (which do depend on input quantities). The fixed cost is **2400 points**.

In each round, your per-unit input costs are **randomly drawn** and **can differ across participants**.

The per-unit cost of Input 1 is equally likely to take any value between 560 and 700 (in steps of 7). The per-unit cost of Input 2 is equally likely to take any value between 80 and 100 (in steps of 1). **These two costs are perfectly linked: in every round, Input 1's unit cost is exactly seven times Input 2's unit cost.**

Your total variable cost is: (unit cost of Input 1 × your quantity of Input 1) + (unit cost of Input 2 × your quantity of Input 2).

Bidding & Interface

- You will place bids using **sliders** on the screen.
- You will see **your own per-unit costs** and the possible net payoffs you would receive for winning the auction.
- You will **not** see your opponent's per-unit costs.

Feedback

After each round:

- The loser sees only their own result.
- The winner sees their own result and a reminder of what the loser's payoff is.

Payment

- At the end of the experiment, one round will be chosen at random to be the round that counts. Your payment will be determined by the number of points you earned in that round.

Next

Figure 30: Negative Correlation Instructions

Instructions: Fixed-Payment Auction

PLEASE READ CAREFULLY AND DO NOT PRESS NEXT UNTIL INSTRUCTED TO DO SO.

Auction Rules

In these auctions, you will use a slider to choose a **fixed payment amount**. This is your **bid**.

- **Lower bids** mean the government pays a smaller amount overall (you receive less).
- The bidder with the **lowest bid** wins. If bids are tied, the winner is chosen at random with equal probability.

What happens if you win?

Your final points for the round are:

- **Your endowment** (10000), plus **your bid**,
- minus, for each input: **cost per unit × actual units**,
- minus the **fixed cost** of 2400 points.

What happens if you lose?

You simply keep your endowment (10000) for that round (no gains or losses from the project).

Click Next to see the instructions for the other type of auction.

Next

Figure 31: Instructions for Cash Auctions

Instructions: Cost-Share Auction

PLEASE READ CAREFULLY AND DO NOT PRESS NEXT UNTIL INSTRUCTED TO DO SO.

Auction Rules

In these auctions, you will use a slider to choose the **percent of variable costs you request to be covered**. This is your **bid**.

- **Lower bids** mean the government covers a *smaller* share (you cover more).
- The bidder who asks for the **smallest share** to be covered wins. If bids are tied, the winner is chosen at random with equal probability.

The winner will receive a baseline payment of **4400 points**.

What happens if you win?

Your final points for the round are:

- **Your endowment** (10000), plus the **baseline payment** (4400),
- minus **your share of the project's variable costs** (based on the inputs actually used),
- minus the **fixed cost** of 2400 points.

The project's variable cost depends on how many units of each input are used in that round (which can be from Set A or Set B). "Your share of the cost" depends on your bid. For example, if you bid 40%, then 40% of the variable costs will be paid by the government and you pay the remaining 60%.

What happens if you lose?

You simply keep your endowment (10000) for that round (no gains or losses from the project).

On the next page, you will find a few examples to help you better understand the rules. Please go through them at your own pace. After you have reviewed the examples, click Next again to begin the quiz.

Next

Figure 32: Instructions for Equity-Bid Auctions

Instructions: Unit-Price Auction

PLEASE READ CAREFULLY AND DO NOT PRESS NEXT UNTIL INSTRUCTED TO DO SO.

Auction Rules

In these auctions, you will use a slider to submit two **unit prices** that you want to be paid for the inputs: one price per unit of Input 1 and one price per unit of Input 2. Your **score** for the auction is the sum of your two unit prices.

- **Lower unit prices** mean the government pays a *smaller* amount per unit (you receive less).
- The bidder with the **lowest score** wins. If bids are tied, the winner is chosen at random with equal probability.

What happens if you win?

Your final points for the round are:

- **Your endowment** (10000),
- plus, for each input: **(your unit price – cost per unit) × actual units**,
- minus the **fixed cost** of 2400 points.

If your price is above cost, you gain on that input; if it's below cost, you lose on that input. The actual units can be from Set A or Set B.

What happens if you lose?

You simply keep your endowment (10000) for that round (no gains or losses from the project).

On the next page, you will find a few examples to help you better understand the rules. Please go through them at your own pace. After you have reviewed the examples, click Next again to begin the quiz.

Next

Figure 33: Instructions for Scaling Auctions

Examples

Please read both examples carefully.

Example 1: Fixed-Payment Auction

- Ann bids **4000** points. Bob bids **4300** points. Ann wins (lowest bid).
- Ann's per-unit costs this round are:
 - Input 1 cost = 360
 - Input 2 cost = 180
- If Set A (2,2) is realized:
 - Variable cost = $2 \times 360 + 2 \times 180 = 1080$
 - Final points = $10000 + 4000 - 1080 - 2400 = \mathbf{10520}$
- If Set B (6,6) is realized:
 - Variable cost = $6 \times 360 + 6 \times 180 = 3240$
 - Final points = $10000 + 4000 - 3240 - 2400 = \mathbf{8360}$
- Bob loses and receives **10000** points.

Example 2: Cost-Share Auction

- Ann bids **30%**. Bob bids **42%**. Ann wins (lowest requested share).
- Ann's per-unit costs: Input 1 = 360, Input 2 = 180.
- Baseline payment in this auction is **4400** points.
- Because Ann bid 30%, she pays 70% of variable cost.
- If Set A (2,2):
 - Variable cost = 1080
 - Final points = $10000 + 4400 - 0.70 \times 1080 - 2400 = \mathbf{11244}$
- If Set B (6,6):
 - Variable cost = 3240
 - Final points = $10000 + 4400 - 0.70 \times 3240 - 2400 = \mathbf{9732}$
- Bob loses and receives **10000** points.

When you are ready, click Next to continue to the quiz.

Next

Figure 34: Examples (Positive Correlation, Equity-Bid Treatment)

Examples

Please read both examples carefully.

Example 1: Fixed-Payment Auction

- Ann bids **3500** points. Bob bids **3800** points. Ann wins (lowest bid).
- Ann's per-unit costs this round are:
 - Input 1 cost = 630
 - Input 2 cost = 90
- If Set A (1,5) is realized:
 - Variable cost = $1 \times 630 + 5 \times 90 = 1080$
 - Final points = $10000 + 3500 - 1080 - 2400 = \mathbf{10020}$
- If Set B (5,1) is realized:
 - Variable cost = $5 \times 630 + 1 \times 90 = 3240$
 - Final points = $10000 + 3500 - 3240 - 2400 = \mathbf{7860}$
- Bob loses and receives **10000** points.

Example 2: Cost-Share Auction

- Ann bids **40%**. Bob bids **46%**. Ann wins (lowest requested share).
- Ann's per-unit costs: Input 1 = 630, Input 2 = 90.
- Baseline payment in this auction is **4400** points.
- Because Ann bid 40%, she pays 60% of variable cost.
- If Set A (1,5):
 - Variable cost = 1080
 - Final points = $10000 + 4400 - 0.60 \times 1080 - 2400 = \mathbf{11352}$
- If Set B (5,1):
 - Variable cost = 3240
 - Final points = $10000 + 4400 - 0.60 \times 3240 - 2400 = \mathbf{10056}$
- Bob loses and receives **10000** points.

When you are ready, click Next to continue to the quiz.

Next

Figure 35: Examples (Negative Correlation, Equity-Bid Treatment)

Examples

Please read both examples carefully.

Example 1: Fixed-Payment Auction

- Ann bids **4000** points. Bob bids **4300** points. Ann wins (lowest bid).
- Ann's per-unit costs this round are:
 - Input 1 cost = 360
 - Input 2 cost = 180
- If Set A (2,2) is realized:
 - Variable cost = $2 \times 360 + 2 \times 180 = 1080$
 - Final points = $10000 + 4000 - 1080 - 2400 = \mathbf{10520}$
- If Set B (6,6) is realized:
 - Variable cost = $6 \times 360 + 6 \times 180 = 3240$
 - Final points = $10000 + 4000 - 3240 - 2400 = \mathbf{8360}$
- Bob loses and receives **10000** points.

Example 2: Unit-Price Auction

- Ann submits unit prices: Input 1 = **760**, Input 2 = **380**.
- Ann's score is $760 + 380 = \mathbf{1140}$. Bob's score is 1160, so Ann wins.
- Ann's per-unit costs: Input 1 = 360, Input 2 = 180.
- If Set A (2,2):
 - Final points = $10000 + [2 \times (760 - 360) + 2 \times (380 - 180)] - 2400 = \mathbf{8800}$
- If Set B (6,6):
 - Final points = $10000 + [6 \times (760 - 360) + 6 \times (380 - 180)] - 2400 = \mathbf{11200}$
- Bob loses and receives **10000** points.

When you are ready, click Next to continue to the quiz.

Next

Figure 36: Examples (Positive Correlation, Scaling Treatment)

Examples

Please read both examples carefully.

Example 1: Fixed-Payment Auction

- Ann bids **3500** points. Bob bids **3800** points. Ann wins (lowest bid).
- Ann's per-unit costs this round are:
 - Input 1 cost = 630
 - Input 2 cost = 90
- If Set A (1,5) is realized:
 - Variable cost = $1 \times 630 + 5 \times 90 = 1080$
 - Final points = $10000 + 3500 - 1080 - 2400 = \mathbf{10020}$
- If Set B (5,1) is realized:
 - Variable cost = $5 \times 630 + 1 \times 90 = 3240$
 - Final points = $10000 + 3500 - 3240 - 2400 = \mathbf{7860}$
- Bob loses and receives **10000** points.

Example 2: Unit-Price Auction

- Ann submits unit prices: Input 1 = **1200**, Input 2 = **171**.
- Ann's score is $1200 + 171 = \mathbf{1371}$. Bob's score is 1400, so Ann wins.
- Ann's per-unit costs: Input 1 = 630, Input 2 = 90.
- If Set A (1,5):
 - Final points = $10000 + [1 \times (1200 - 630) + 5 \times (171 - 90)] - 2400 = \mathbf{8575}$
- If Set B (5,1):
 - Final points = $10000 + [5 \times (1200 - 630) + 1 \times (171 - 90)] - 2400 = \mathbf{10531}$
- Bob loses and receives **10000** points.

When you are ready, click Next to continue to the quiz.

Next

Figure 37: Examples (Negative Correlation, Scaling Treatment)

Quiz

You will now be given a series of questions to check your understanding of the instructions and examples. You will be paid \$1 for each answer you get correct.

At the end of the experiment, how is your final payment determined?

- ☐ By summing your total points across all rounds
- ☐ By averaging your points across all rounds
- ☐ By selecting one round at random and paying based on its points
- ☐ By paying based on a round you won the auction

Which of the following is true about the project's input requirements?

- ☐ You know whether Set A or Set B will be realized before bidding
- ☐ Set A and Set B are equally likely, and the realized set is unknown when bidding
- ☐ Set A occurs with 75% probability and Set B with 25%
- ☐ The realized set is chosen by the winner

What happens if you lose an auction round?

- ☐ You pay only the fixed cost
- ☐ You get your endowment only
- ☐ You receive an extra payment
- ☐ You must skip the next round

In the Cost-Share auction, what do you submit?

- ☐ Two unit prices, one per input
- ☐ A fixed cost level
- ☐ A lump-sum payment request
- ☐ A percent of variable costs you request to be covered

In the Cost-Share auction, how is the winner determined?

- ☐ Smallest requested share to be covered
- ☐ Bid closest to 50%
- ☐ Highest requested share to be covered
- ☐ Highest baseline payment

When you believe you have answered all questions correctly, press Next to check your answers.

Next

Figure 38: Quiz (Equity-Bid Treatment)

Quiz

You will now be given a series of questions to check your understanding of the instructions and examples. You will be paid \$1 for each answer you get correct.

At the end of the experiment, how is your final payment determined?

- ☐ By summing your total points across all rounds
- ☐ By averaging your points across all rounds
- ☐ By selecting one round at random and paying based on its points
- ☐ By paying based on a round you won the auction

Which of the following is true about the project's input requirements?

- ☐ You know whether Set A or Set B will be realized before bidding
- ☐ Set A and Set B are equally likely, and the realized set is unknown when bidding
- ☐ Set A occurs with 75% probability and Set B with 25%
- ☐ The realized set is chosen by the winner

What happens if you lose an auction round?

- ☐ You pay only the fixed cost
- ☐ You get your endowment only
- ☐ You receive an extra payment
- ☐ You must skip the next round

In the Unit-Price auction, what do you submit?

- ☐ A single lump-sum payment request
- ☐ Only a price for Input 1
- ☐ A percent of variable costs to be covered
- ☐ Two unit prices: one for Input 1 and one for Input 2

In the Unit-Price auction, how is the winner determined?

- ☐ Lowest score (price for Input 1 + price for Input 2)
- ☐ Lowest price for Input 1 only
- ☐ Lowest fixed cost
- ☐ Highest total unit-price sum

When you believe you have answered all questions correctly, press Next to check your answers.

Next

Figure 39: Quiz (Scaling Treatment)

Quiz Answers

The answers for the quiz are given below. Please review the answers and note any mistakes you have made.

Question 1: At the end of the experiment, how is your final payment determined?

Correct Answer: By selecting one round at random and paying based on its points

Your Answer: By averaging your points across all rounds

Question 2: Which of the following is true about the project's input requirements?

Correct Answer: Set A and Set B are equally likely, and the realized set is unknown when bidding

Your Answer: Set A occurs with 75% probability and Set B with 25%

Question 3: What happens if you lose an auction round?

Correct Answer: You get your endowment only

Your Answer: You receive an extra payment

Question 4: In the Cost-Share auction, what do you submit?

Correct Answer: A percent of variable costs you request to be covered

Your Answer: A percent of variable costs you request to be covered

Question 5: In the Cost-Share auction, how is the winner determined?

Correct Answer: Smallest requested share to be covered

Your Answer: Highest requested share to be covered

You earned \$1.0 from your correct answers. Please review any questions you answered incorrectly. When you are ready to begin the first round, click the next button.

Next

Figure 40: Quiz Answers (Equity-Bid Treatment)

Quiz Answers

The answers for the quiz are given below. Please review the answers and note any mistakes you have made.

Question 1: At the end of the experiment, how is your final payment determined?

Correct Answer: By selecting one round at random and paying based on its points

Your Answer: By selecting one round at random and paying based on its points

Question 2: Which of the following is true about the project's input requirements?

Correct Answer: Set A and Set B are equally likely, and the realized set is unknown when bidding

Your Answer: Set A and Set B are equally likely, and the realized set is unknown when bidding

Question 3: What happens if you lose an auction round?

Correct Answer: You get your endowment only

Your Answer: You pay only the fixed cost

Question 4: In the Unit-Price auction, what do you submit?

Correct Answer: Two unit prices: one for Input 1 and one for Input 2

Your Answer: Two unit prices: one for Input 1 and one for Input 2

Question 5: In the Unit-Price auction, how is the winner determined?

Correct Answer: Lowest score (price for Input 1 + price for Input 2)

Your Answer: Lowest score (price for Input 1 + price for Input 2)

You earned \$4.0 from your correct answers. Please review any questions you answered incorrectly. When you are ready to begin the first round, click the next button.

Next

Figure 41: Quiz Answers (Scaling Treatment)

Round 1: Fixed-Payment Auction

Each participant is assigned **different per-unit costs** for the inputs in each round. You see only your own per-unit costs. Your opponent's are hidden.

In this round, your per-unit costs are **376 for Input 1** and **188 for Input 2**.

Use the slider to choose a **payment** you request if you win the auction. The bidder asking for the **lowest payment** wins.

If you win: $10000 + \text{your requested payment} - \text{your per-unit costs} \times \text{units} - 2400$.

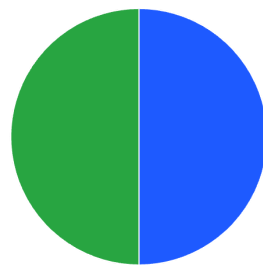
If you lose: you keep 10000.

0

10000

Click Next when you're ready to submit your bid.

Next



■ chance of input set B
■ chance of input set A



Key Reminders

- Only **one randomly selected round** will be paid in cash.
- Your endowment is **10000 points**.
- Conversion rate: **200 points = \$1**.
- Fixed cost if you win: **2400 points**.
- Input requirements: Input 1 **Set A = 2, Set B = 6**; Input 2 **Set A = 2, Set B = 6**.
- Personal per-unit costs: Your input prices are **private** and **different for each bidder**; each round, Input 1 is randomly drawn from 320 to 400 and Input 2 from 160 to 200, with all values in each range equally likely, and Input 1 is always exactly double Input 2.

Figure 42: Cash Auction Bidding Page

Round 1: Fixed-Payment Auction

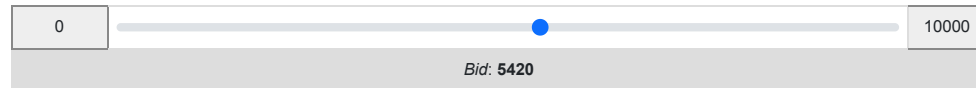
Each participant is assigned **different per-unit costs** for the inputs in each round. You see only your own per-unit costs. Your opponent's are hidden.

In this round, your per-unit costs are **376 for Input 1** and **188 for Input 2**.

Use the slider to choose a **payment** you request if you win the auction. The bidder asking for the **lowest payment** wins.

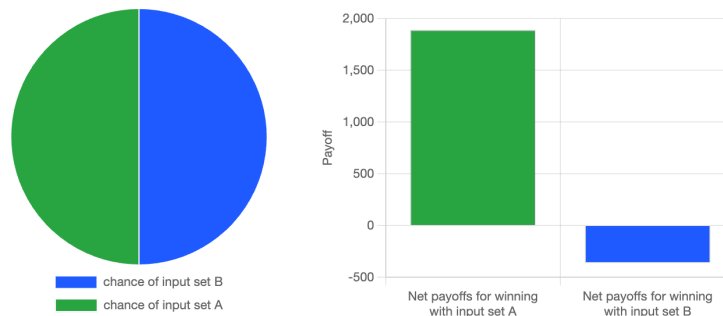
If you win: $10000 + \text{your requested payment} - \text{your per-unit costs} \times \text{units} - 2400$.

If you lose: you keep 10000.



Click Next when you're ready to submit your bid.

Next



Key Reminders

- Only **one randomly selected round** will be paid in cash.
- Your endowment is **10000 points**.
- Conversion rate: **200 points = \$1**.
- Fixed cost if you win: **2400 points**.
- Input requirements: Input 1 **Set A = 2, Set B = 6**; Input 2 **Set A = 2, Set B = 6**.
- Personal per-unit costs: Your input prices are **private** and **different for each bidder**; each round, Input 1 is randomly drawn from 320 to 400 and Input 2 from 160 to 200, with all values in each range equally likely, and Input 1 is always exactly double Input 2.

Figure 43: Cash Auction Bidding Page (tool active)

Round 1 Results

You did not win this round.

Your bid was 7080 and the other bid was 5420.

Thus, your payoff for this round will be 10000.

Next

Figure 44: Results Page (Loss)

Round 1 Results

Congratulations! You won the auction.

Your bid was 5420 and the other bid was 7080.

Your payoff for this round will be $10000 + 5420 - (376 \times 6 + 188 \times 6) - 2400 = \mathbf{9636}$ with one-half chance and $10000 + 5420 - (376 \times 2 + 188 \times 2) - 2400 = \mathbf{11892}$ with one-half chance.

The payoff of the bidder that did not win will be 10000 with certainty.

Next

Figure 45: Cash Auction Results Page

Round 2: Cost-Share Auction

Each participant is assigned **different per-unit costs** for the inputs in each round. You see only your own per-unit costs. Your opponent's are hidden.

In this round, your per-unit costs are **392 for Input 1** and **196 for Input 2**.

Use the slider to choose the **percent of variable costs you request to be covered**. The bidder asking for the **smallest share** wins.

If you win: $10000 + \text{baseline payment } (4400) - (1 - \text{your bid}) \times \text{your per-unit costs} \times \text{units} - 2400$.

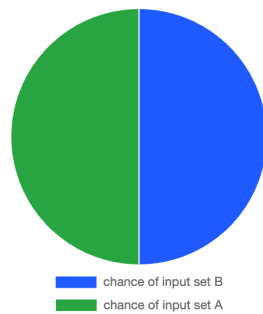
If you lose: you keep 10000.

0

50

Click Next when you're ready to submit your bid.

Next



Key Reminders

- Only **one randomly selected round** will be paid in cash.
- Your endowment is **10000 points**.
- Conversion rate: **200 points = \$1**.
- Fixed cost if you win: **2400 points**.
- Input requirements: Input 1 **Set A = 2, Set B = 6**; Input 2 **Set A = 2, Set B = 6**.
- Personal per-unit costs: Your input prices are **private** and **different for each bidder**; each round, Input 1 is randomly drawn from 320 to 400 and Input 2 from 160 to 200, with all values in each range equally likely, and Input 1 is always exactly double Input 2.

Figure 46: Equity-Bid Auction Bidding Page

Round 2: Unit-Price Auction

Each participant is assigned **different per-unit costs** for the inputs in each round. You see only your own per-unit costs. Your opponent's are hidden.

In this round, your per-unit costs are **400 for Input 1** and **200 for Input 2**.

Set your **bid per unit** for Input 1 and Input 2 using the sliders. These bids will be added together to calculate your **score**. The **lowest score wins**.

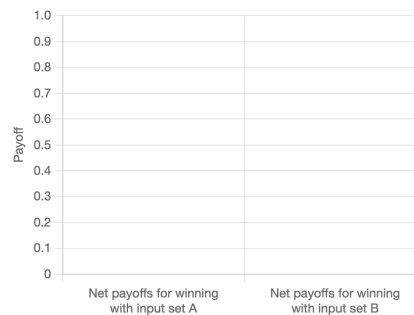
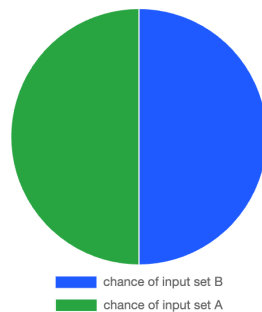
If you win: $10000 + (\text{your bid} - \text{your per-unit costs}) \times \text{units} - 2400$.

If you lose: you keep 10000.

0		1600
0		1600

Click Next when you're ready to submit your bid.

Next



Key Reminders

- Only **one randomly selected round** will be paid in cash.
- Your endowment is **10000 points**.
- Conversion rate: **200 points = \$1**.
- Fixed cost if you win: **2400 points**.
- Input requirements: Input 1 **Set A = 2, Set B = 6**; Input 2 **Set A = 2, Set B = 6**.
- Personal per-unit costs: Your input prices are **private** and **different for each bidder**; each round, Input 1 is randomly drawn from 320 to 400 and Input 2 from 160 to 200, with all values in each range equally likely, and Input 1 is always exactly double Input 2.

Figure 47: Scaling Auction Bidding Page

Round 2 Results

Congratulations! You won the auction.

Your bid was 12% and the other bid was 34%.

Your payoff for this round will be $10000 + 4400 - (1 - (12/100)) \times ((332 \times 6) + (166 \times 6)) - 2400 = \mathbf{9371 \text{ with one-half chance}}$ and $10000 + 4400 - (1 - (12/100)) \times ((332 \times 2) + (166 \times 2)) - 2400 = \mathbf{11124 \text{ with one-half chance}}$.

The payoff of the bidder that did not win will be 10000 with certainty.

Next

Figure 48: Equity-Bid Auction Results Page

Round 2 Results

Congratulations! You won the auction.

Your score was 1680.0 and the other bidder's score was 2720.0.

Your payoff for this round will be $10000 + (1060 - 400) \times 6 + (620 - 200) \times 6 - 2400 = \mathbf{14080 \text{ with one-half chance}}$ and $10000 + (1060 - 400) \times 2 + (620 - 200) \times 2 - 2400 = \mathbf{9760 \text{ with one-half chance}}$.

The payoff of the bidder that did not win will be 10000 with certainty.

Next

Figure 49: Scaling Auction Results Page

Payoffs

That was the last match of the experiment. The computer has randomly selected match 1 to be the match that counts for your payment.

In that round, you won the auction. You ended up needing the Set B inputs for the project, so your payoff in points for that match was 9636. This corresponds to \$48.18.

With your quiz payment of \$0.0, this gives a total of \$48.18.

The next few pages consist of survey questions. Please complete the survey while we prepare your payment.

Next

Figure 50: Final Payoffs

Survey

Please answer the following questions.

What is your age?

What gender do you identify with the most?

- ☐ Female
☐ Male
☐ Other/Prefer Not to Say

Is English your first language?

- ☐ Yes
☐ No

Are you completing or have you completed an economics degree?

- ☐ Yes
☐ No

Next

Figure 51: Demographic Survey

Survey

Please answer the following questions.

A bat and a ball cost 22 dollars in total. The bat costs 20 dollars more than the ball. How many dollars does the ball cost?

"If it takes 5 machines 5 minutes to make 5 widgets, how many minutes would it take 100 machines to make 100 widgets?"

In a lake, there is a patch of lily pads. Every day, the patch doubles in size. If it takes 48 days for the patch to cover the entire lake, how many days would it take for the patch to cover half of the lake?

Next

Figure 52: Cognitive Reflection Test

Survey

Please answer the following questions:

The instructions of this study were easy to understand.

☐ Strongly disagree ☐ Disagree ☐ Neither agree nor disagree ☐ Agree ☐ Strongly agree

What, if anything, were you confused about in the study?

I knew how to make the decisions that were best for me in the experiment.

☐ Strongly disagree ☐ Disagree ☐ Neither agree nor disagree ☐ Agree ☐ Strongly agree

How did you make decisions in the study?

Of the two types of auctions today, I preferred to participate in the Cost-Share auction.

☐ Strongly disagree ☐ Disagree ☐ Neither agree nor disagree ☐ Agree ☐ Strongly agree

Please briefly explain that response.

What do you think this study was about?

Next

Figure 53: Experiment Feedback